



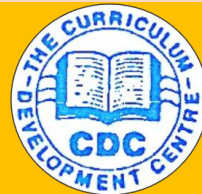
Republic of Zambia

Ministry of Education
DIRECTORATE OF CURRICULUM DEVELOPMENT

AGRICULTURAL SCIENCE

TEACHING MODULE

FORM 2 TERM 1 & 2



Developed by the Curriculum Development Centre
LUSAKA
2026

COPYRIGHT

© Curriculum Development Centre 2026.

All rights reserved. No parts of this publication may be reproduced, stored in a retrieval system or transmitted by electronic, mechanical, photocopying, recording or other means, without the prior permission of the Curriculum Development Centre

AUTHORS

Brian Mwitwa Chimbanga

Lecturer, Natural Resources Development College, Lusaka

Godfrey Kayamba

Head Teacher, Chavuma Boarding Secondary School, Chavuma

George Suwilanji Simuyemba

Lecturer, Kasama College of Education, Kasama (ASEDAZ)

COORDINATORS

Martha Lisulo

Curriculum Development Specialist-Biological Sciences, Curriculum Development Centre, Lusaka

Kenneth Mudenda

Senior Curriculum Specialist –Natural Sciences, Curriculum Development Centre, Lusaka

Samson Njapau

Principal Curriculum Specialist - Natural Sciences - Curriculum Development Centre, Lusaka

VISION

Quality, life-long education for all which is accessible, inclusive and relevant to individual, national and global needs which align with the 8 NDP and SDG 4.

Contents

COPYRIGHT	I
VISION.....	III
PREFACE	VI
ACKNOWLEDGEMENTS	VII
INTRODUCTION	VIII
PURPOSE OF THE MODULE.....	IX
HOW TO USE THE MODULE	X
ICONS USED IN THIS MODULE	XI
TERM 1	1
TOPIC 2.1.0: AGRICULTURE IN ZAMBIA.....	1
SUB-TOPIC 2.1.1 LAND TENURE AND LAND USE	3
TOPIC 2.2.0 SOIL SCIENCE	14
SUB-TOPIC 2.2.1 SOIL FERTILITY (NUTRIENT CYCLES)	16
TOPIC 2.3.0. CLIMATE - SMART AGRICULTURE	22
SUB-TOPIC 2.3.1. CLIMATE CHANGE IMPACT ON AGRICULTURE AND FOOD SECURITY	24
TOPIC 2.4.0 FARM STRUCTURES AND MACHINERY	32
SUB-TOPIC 2.4.1 FARM BUILDINGS.....	33
SUBTOPIC 2.4.2. ANIMAL AND TRACTOR DRAWN IMPLEMENTS.....	38
TERM 2	44
TOPIC 2.0 CROP PRODUCTION	45
SUB-TOPIC 2.5.1 EXTERNAL AND INTERNAL MORPHOLOGY OF PLANTS.....	47
SUB-TOPIC 2.5.2 FIELD CROP PRODUCTION.....	62

TOPIC 2.6.0 LIVESTOCK PRODUCTION.....	81
SUB-TOPIC 2.6.1 POULTRY PRODUCTION IN ZAMBIA.....	84
BIBLIOGRAPHY.....	119

PREFACE

The Agricultural Science Ordinary Level Teaching Module has been developed to provide guidance to the teacher on how to deliver competence-based lessons which are age-appropriate for learners at Form 2. The teaching module is aimed at providing quality education that is aligned with the Competence-Based Curriculum and 21st Century Skills. It also aims to facilitate the building of knowledge, skills, values and positive attitudes that are aimed at enabling children to live and grow into productive and useful members of their communities and the Zambian society.

This teaching module is interconnected to the Form one module. The guide provided in this teaching module is aimed at facilitating the acquisition of knowledge, competences and values to nurture a learner who is creative, critical and an analytical thinker. The suggested learning activities are designed to offer learners hands-on and minds-on experiences through manipulation of real objects, interaction with nature and learning through activities. Thus, the learner will develop knowledge, desirable lifelong skills, values and positive attitudes needed for their personal and national development.

It is hoped that the guidance provided in this module will make learning of Agricultural Science at Ordinary level more meaningful and enjoyable as it is highly activity oriented and allows for a smooth transition into the Constructivist way of learning.

Kelvin Mambwe (PhD.)
Permanent Secretary- Educational Services
MINISTRY OF EDUCATION

ACKNOWLEDGEMENT

This Agricultural Science Ordinary Level Teaching Module incorporates guidance on how to deliver Competence Based lessons in this field of Science. The module is a build-up of topics introduced at Form one (1).

Many thanks go to individuals, institutions and organizations that participated in the successful development of this Module. These include Teachers, Agricultural Science Educators Association of Zambia (ASEDAZ) Lecturers from colleges in Zambia and experts from Directorates in the Ministry of Education.

I also recognise the commitment and hard work of Staff at the Curriculum Development Centre in ensuring that this syllabus is successful.

Charles Ndakala (Dr.)
Director- Curriculum Development
MINISTRY OF EDUCATION

INTRODUCTION

The Agricultural Science Ordinary Level Teaching Module has been developed to guide teachers on how to facilitate competence-based lessons at this level. The module aims to provide quality education that is aligned with the competence-based curriculum and 21st Century Skills, that is developing knowledge, skills, values and positive attitudes that are aimed at enabling Learners to live and grow into productive and useful members of their communities and the Zambian society. The syllabus presents Agricultural Science as an applied science. Agricultural science will be taught using learner centered activities which will help learners to acquire desirable lifelong skills, values and positive attitudes needed for their personal and national development.

Agricultural science will be taught using learner centred activities, which will help learners to acquire desirable lifelong skills, values and positive attitudes needed for their personal and national development. Agricultural Science, as an applied science, will be taught in a practical manner, in which learners will be involved in carrying out agricultural activities, which will include growing crops, rearing poultry, keeping livestock, apiculture and aquaculture. Therefore, Agricultural Science activities will be done both in the laboratory where experiments will be conducted and, in the field, where field trials/observations will be done. The assessment in the teaching module is part of the school-based assessment. It will be emphasized by using various methods of testing according to sub-topics as part of the normal teaching and learning process and therefore, the assessment opportunities will also occur during this normal process.

The assessment to measure each Specific Competence will occur in three forms:

- Observation – watching learners working (good for assessing skills and values)
- Conversation – asking questions and talking to learners (good for assessing knowledge and understanding)
- Product – appraising the learners' work (writing reports, translation, calculations, presentations, maps, diagrams, models etc.) in this context the product is seen as something physical and permanent that the teacher can keep and look at.

PURPOSE OF THE MODULE

The purpose of the module is to illustrate teaching and learning activities that:

- i. are consistent with teaching and learning in a competence-based curriculum,
- ii. show processes that allow achievement of 21st Century Skills such as creativity, collaboration, communication and critical thinking,
- iii. illustrate what activity-based and learner-centred teaching and learning could look like.

HOW TO USE THE MODULE

This module provides suggested learning activities and teaching strategies that enhance learners' scientific competences. The suggested activities are designed to offer learners hands-on experiences through manipulation of real objects, interaction with nature and learning through practice, bearing in mind that the desired competences and 21st century skills will NOT be achieved by the topics 'covered' but how the teaching and learning activities are carried out.

Activities in the Module are meant to be incorporated in the teacher's lesson plan. Thus, the activities are NOT meant to replace the teacher's lesson plan.

The topics and activities in the Module are as specified in the 2024 Agricultural Science syllabus. There might be terms used in the Module that are not the same as those in the syllabus, but this is in the interest of using language that might be more appropriate for the learner. Teachers are encouraged to be mindful as well about what language is within the grasp of their learners and not use terminology simply because it appears in the syllabus.

Teachers are encouraged to start topics with a hook that provokes their learners' curiosity and motivation. A Hook is an opening statement at the start of the topic that seeks to stimulate learners' curiosity and motivate them to learn. A hook can be a provocative question, a scenario or something that puts learners into a thinking mood. Let the designed hook incline towards problem-solving scenarios.

ICONS USED IN THIS MODULE



General Competences



Introductions



Study skills



Reflections



Summary



Assignments



Terminology



Assessment



Tips



Activity



Case study



Discussion

TERM 1

TOPIC 2.1.0: AGRICULTURE IN ZAMBIA



Introduction

This topic focuses on land tenure and land use. Learners will explore existing land tenure systems and analyze the effects of land tenure and human population growth on agricultural development. In addition, learners will investigate the factors that limit land use in different regions of Zambia and explore the various forms of land use practiced in the country.

Hook: *“Do you know who owns land in your community and do you know that you can own land for 99 years?”*



General Competences

- Analytical thinking,
- Communication,
- Creativity and innovation,
- Collaboration,
- Critical thinking,
- Environmental sustainability and
- Problem solving.



Key Concepts / Processes / Terms

Customary land -	land that is owned and managed according to traditional customs and practices of a local community rather than through formal government land titles or statutory law.
Commercial farming -	type of farming where crops and livestock are produced for sale in the market to make profit rather than for family consumption.
Freehold -	a form of land ownership where a person owns the land permanently and has full legal rights to use, sell, lease or transfer it to heirs.
Land tenure -	is the condition or period under which an occupier holds land.
Land use -	the management and modification of natural land resources for human activities such as farming, housing or industry.
Lease hold-	A form of property ownership where a person (tenant or leaseholder) has the right to use and occupy land or building for a fixed period.
State land -	Land owned and controlled by the state or government.
Subsistence farming-	Type of farming where crops and livestock are kept mainly for family consumption and not for sale.
Traditional leaders-	Individuals such as chiefs, headmen, village leaders who govern and guide communities according to customs, traditions and cultural norms.

Sub-Topic 2.1.1 Land Tenure and Land Use



Introduction

In this sub-topic, learners are going to investigate land tenure systems and land use in Zambia.

Specific Competence

Interpret the land tenure systems and land use in Zambia.

Learning Activities

The following are some of the activities you can use to teach the learners about the land tenure systems and land use in Zambia.

1.1.1 Describing land tenure and land use

Learning materials: Textbooks, Notebook, manilla paper, digital and print media.



Activity 1: Describing land tenure and land use

Procedure

- i. Divide learners into small groups of 3-4
- ii. Ask each group to discuss the terms; land tenure and land use.
- iii. Ask each group to brainstorm on who owns land in their community and how land is used in the school or homes.
- iv. Ask each group to take notes and write their findings on the manilla papers.
- v. Ask each group to present their findings.

Supporting Activities

- i. Viewing video clips on land tenure and land use.

- ii. Inviting officers from Ministry of Lands or traditional leaders.

1.1.2 Exploring existing land tenure systems (Communal, customary, leasehold and state land ...)

Learning Materials: Maps of Zambia showing land use and land ownership, Charts on land tenure systems, Pictures of different land uses, atlases and textbooks.



Activity 1: Exploring existing land tenure systems (Communal, customary, leasehold and state land ...)

Procedure:

- i. Divide learners into small groups of 3-4.
- ii. Ask learners to use maps, charts, pictures and textbooks related to land tenure and land use as provided by the teacher. Learners to identify and list the main land tenure systems in Zambia.
- iii. Ask learners to discuss how each land tenure system influences land use (e.g., grazing, farming, settlement).
- iv. Ask each group to prepare short description of their findings and make presentations.

1.1.2 Exploring existing land tenure systems (Communal, customary, leasehold and state land ...)

Learning Materials: Charts/Posters showing types of land tenure systems, Maps of Zambia indicating land ownership and use, Flash cards, Picture/photographs, Case study handouts, Chalkboard, Work cards/sheets for group work.



Activity 2. Exploring existing land tenure systems (Communal, customary, leasehold and state land...)

Procedure:

- i. Divide learners into small groups of 3-4.
- ii. Assign each group with one land tenure system.
- iii. Ask learners to discuss the assigned land tenure system (focusing on meaning, ownership and uses).
- iv. Ask each group to prepare poster or written summary of land tenure assigned.
- v. Ask learners to present their findings.

1.1.2 Exploring existing land tenure systems (Communal, customary, leasehold and state land ...)



Activity 3: Study tour

Procedure:

- i. Divide learners into small groups of 3-4 pupils.
- ii. Ask learners to visit a nearby community to investigate the system of land tenure in the locality and how the system influences agricultural productivity.
- iii. Ask groups to present findings in class.



Content Tip:

- This procedure promotes collaboration, research skills, communication and understanding of land tenure systems and land uses.

Supporting Activities

- i. Role play
- ii. Viewing video clips on land tenure systems

1.1.3 Analysing the effects of land tenure and human population on agriculture

Learning Materials: Notebooks, questionnaires



Activity 1. Analysing the effects of land tenure and human population on agriculture

Procedure:

- i. Divide learners into small groups of 3-4 pupils.
- ii. Ask learners to visit local farmers in the nearby community. Learners should ask farmers whether they own the land they are using for production.
- iii. Ask learners to find out from farmers the challenges they face in relation to use of land with an increase in human population.
- iv. Ask groups to present findings in class.

1.1.3 Analysing the effects of land tenure and human population on agriculture



Activity 2: Scenario

A village has experienced rapid population growth. Land that once had large farms is now divided into smaller ones.

- i. Put learners into small groups of 3-4 pupils.
- ii. Ask learners to discuss the following questions in relation to the scenario presented above;
 - What problems may arise for farmers due to rapid population growth?
 - How does this affect food production in the community?
 - Suggest solutions that could help mitigate these challenges?
- iii. Ask learners to present their findings.



Content Tip

- Overcultivation
- Land fragmentation
- Reduced productivity

Supporting Activities

- i. Viewing video clips of effects of land tenure and human population on agriculture

1.1.4 Investigating factors that limit the use of land in different areas of Zambia

Learning Materials: Notebooks, Map of Zambia



Activity 1. Investigating factors that limit the use of land in different areas of Zambia

Procedure

- i. Put learners into small groups of 3-4.
- ii. Ask learners to visit a nearby farming area or open land.
- iii. Ask learners to identify factors that limit the use of land in different areas visited.
- iv. Ask learners to report their findings.



Content Tips

- Poor soils, steep slopes, floods, lack of water, drought, population, soil fertility, pests, diseases, land policies, wildlife, thick forest as factors that limit use of land.

- Flood plains such as the Barotse Floodplain limit permanent farming because of seasonal floods.
- Conservation areas like South Luangwa National Park restrict agriculture to protect wildlife.

Supporting material

- **Consulting print and digital resources**

1.1.5 Classifying different forms of land use based on: (a)Agricultural systems (Agroforestry, crop production, livestock production and hydroponics...) (b) Farming practices (Rotations, intercropping, mixed farming and monoculture...)

Learning materials: Charts, pictures, textbooks



Activity 1: Classifying different forms of land use based on: (a)Agricultural systems (Agroforestry, crop production, livestock production and hydroponics...) (b) Farming practices (Rotations, intercropping, mixed farming and monoculture...)

Procedure

- i. Divide learners into small groups of 4-5.
- ii. Ask learners to study the pictures in **Figure 1.0** below.

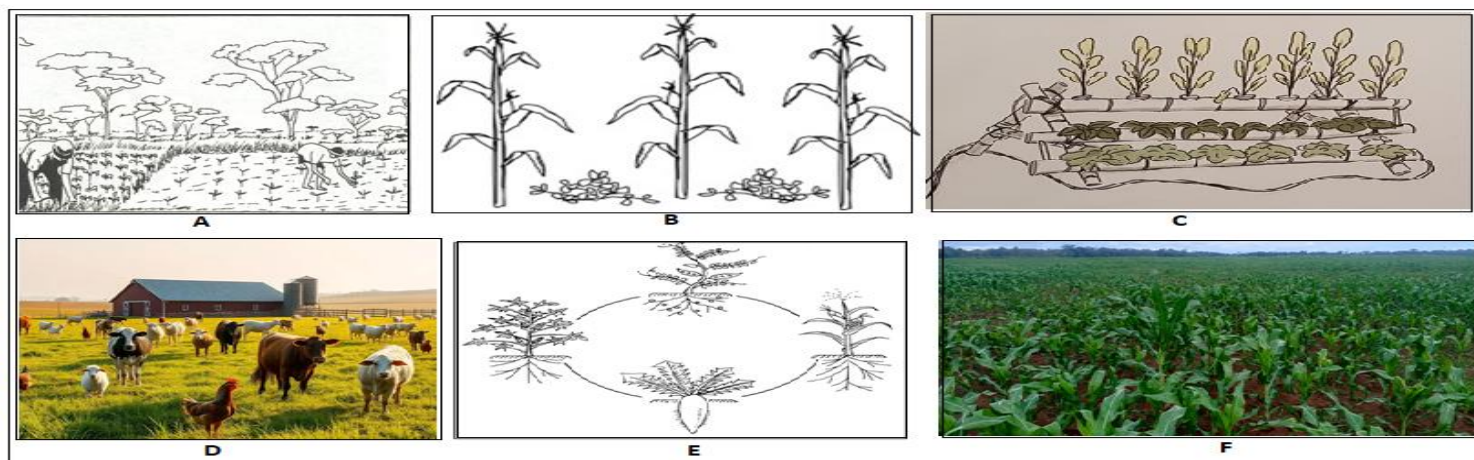


Figure 1.0

- iii. Ask learners to classify the different agricultural systems and farming practices from the illustrations above.
- iv. Ask learners to discuss and report their findings

Supporting Activities

- i. Viewing a video clip with similar activities
- ii. Using electronic or printed materials

1.1.5 Classifying different forms of land use based on: (a)Agricultural systems (Agroforestry, crop production, livestock production and hydroponics...) (b) Farming practices (Rotations, intercropping, mixed farming and monoculture...)



Activity 2: The Scenario Story

- i. Divide learners into small groups of 4-5.
- ii. Ask learners to read the story presented below.

Pupils at Chabila Secondary School were invited by the local agricultural extension officer to learn how people in the area use their land. During the visit, pupils observed the following situations;

- *Mrs. Kapata grows maize and beans on the same field at the same time. On the edges of the field, there are a few cattle grazing in a fenced area.*
- *Mr. Michello has a small garden where tomatoes and rape are grown in plastic bottles filled with water and nutrients without soil used.*
- *Mrs. Mutale has a large open pasture field where many cattle, sheep and goats graze freely on grass as no crops are grown in this area.*
- *Mr. Kaleo grows maize under tall trees such as musangu and acacia. The trees provide shade, and the leaves fall on the soil to make it fertile.*
- *Mr. Kalusa owns a large piece of land, only one crop soya bean is grown season after season.*
- *Ms. Kayombo has subdivided his field into small plots with crops grown in a planned sequence i.e., maize-beans-sweet potatoes-cassava.*

iii. Ask learners to complete the table below by identifying agricultural systems and farming practices demonstrated by different farmers in the scenario given above.

Scenario	Agricultural System	Farming practices	Evidence from the Story
a) Mrs. Kapata	Mixed farming	-	Crops and animals on same farm
b) Mrs. Mutale	-	Monoculture	One crop grown on large land
c)			
d)			
e)			
f)			
g)			

iv. Ask learners to discuss the following questions in their respective groups.

- Why is it that some farmers combine crops and animals.

- Why some farmers grow only one crop.
- Which system or practice is common in their community?
- Ask learners to present their findings.

Supporting Activities

- i. Consult digital resources and print media



Skills Developed

- **Communicating** – Learners sharing their findings with peers and sharing descriptions of land tenure and land use.
- **Collaborating** – Learners working together in groups to explore existing land tenure systems in Zambia.
- **Analytically Thinking** – Learners analysing the effects of land tenure and human population on agriculture.
- **Problem Solving** – Understanding factors that limit the use of land in different areas in order to create possible solutions for proper use of land.
- **Environmental Sustainability** – Adopting suitable agricultural systems and farming practices that are environmental user friendly for improving productivity in a sustainable manner.

Expected Standard

Land tenure systems and land use in Zambia interpreted correctly.



Assessment

The tasks in this section are just samples. A teacher is free to give own assessment activities.

Task 1

In Isoko Village, community members engage in various economic activities such as growing vegetables and field crops, rearing poultry, piggery, cattle, goats, and other livestock. Recently, the exploration and discovery of valuable mineral resources in the area has increased demand for land. As a result, there is growing competition for land among mining companies, farmers, government institutions, and potential investors planning development projects. Due to this situation, the local leadership has organized a community sensitization program to help residents understand how land should be owned, managed, and used responsibly. As a learner who has studied land management concepts, you have been invited to participate in educating the community.

Prepare a short presentation for the community sensitization program addressing the following:

- i. Explain the meaning of term ‘land tenure’;
- ii. Describe the main types of land tenure systems practiced in Zambia;
- iii. Outline the key characteristics of each land tenure system, and
- iv. Discuss how proper understanding of land tenure and land use can help reduce conflicts and promote sustainable development in the community.

Task 2

The Ministry of Lands and Natural Resources has organized an inter-school paper presentation aimed at improving learners’ understanding of land management and ownership systems in Zambia. The event will bring together learners from different schools to present and discuss important issues related to land administration and sustainable land use. As a learner who has demonstrated strong knowledge in land tenure and land use, your school has selected you to represent it at this event. You are required to prepare a paper presentation that will help fellow learners understand how land is owned, accessed, and managed in Zambia.

Prepare a well-structured presentation in which you:

- i. Explain the meaning of term ‘land use’;
- ii. Identify and describe the main types of land tenure systems practiced in Zambia;

- iii. Discuss how human population growth influences agricultural land use, and
- iv. Discuss the advantages of each land tenure system in promoting land security and sustainable development.

Task 3

The District Agricultural Coordinating Officer has organized a District Conference on Land Use aimed at discussing challenges and opportunities related to land management and agricultural development in local communities. The conference will bring together farmers, agricultural officers, community leaders, and learners to share knowledge and propose solutions for improving land utilization. As a learner with wide knowledge of land tenure, the effects of human population on agriculture, factors that limit land use in Zambia, and different forms of land use based on agricultural systems and farming practices, you have been selected to prepare a report that will be presented during the conference.

Prepare a well-organized report for presentation at the conference in which you:

- i. Identify and explain the major factors that limit land use in Zambia;
- ii. Describe the different forms of land use based on agricultural systems and farming practices;
- iii. Suggest possible measures that communities can adopt to promote sustainable and efficient use of land, and
- iv. Analyse different forms of land use based on agricultural systems and farming practices.



Sub-Topic 2.1.1 Summary

In Zambia, land tenure systems determine how land is owned and managed. The main types are customary, communal, state land, freehold and leasehold land ownership. Understanding land tenure and land use helps plan agriculture and manage resources effectively

TOPIC 2.2.0 SOIL SCIENCE



Introduction

This sub-topic covers soil fertility. Soil fertility is the ability of soil to provide essential nutrients to plants for healthy growth. Fertile soil contains the right balance of nitrogen, phosphorus, potassium and other minerals. Learners will engage in demonstrating understanding of the nutrient cycles through analysing and illustrating nitrogen and carbon cycles in relation to soil fertility.



General Competencies

- Analytical thinking
- Communication
- Creativity and innovation
- Collaboration
- Critical thinking
- Problem solving
- Environmental sustainability

Hook: *“Do you know that even if soil may look ordinary, it is hidden treasure full of nutrients, water and life. Today, we will uncover its secrets.”*



Key Concepts/Processes/Terms

- Ammonification** - is the process by which microorganisms (bacteria and fungi) break down dead plants, animals and waste materials' convert organic nitrogen into ammonia (NH_3) or ammonium (NH_4^+) in the soil.
- Denitrification** - is the process in which bacteria convert nitrates (NO_3^-) in the soil into nitrogen gas (N_2), which is then released back into the atmosphere.
- Fossilization** - is the process by which the remains or traces of plants and animals from the past are preserved in rocks, forming fossils over a long period of time.
- Organic matter** - Decomposed plant and animal materials in the soil that improves fertility.
- Nitrogen fixation** - the process by which nitrogen from the air is converted into forms usable by plants often by bacteria.
- Nitrification** - is the conversion of ammonia or ammonium into nitrites (NO_2^-) and then into nitrates (NO_3^-) by nitrifying bacteria in the soil.
- Compost manure** - is organic fertiliser made from decomposed plant materials, food waste and animal waste which is added to soil to improve its fertility and structure.
- Soil fertility** - the capacity of soil to provide essential nutrients in adequate amounts for plant growth.

Sub-Topic 2.2.1 Soil Fertility (Nutrient Cycles)



Introduction

Soil fertility is the ability of soil to supply nutrients required for plant growth. Nutrient cycles involve the continuous recycling of nutrients between the soil, plants, animals and the environment, helping maintain soil productivity. In this subtopic, learners will appreciate the importance of nutrients for sustaining plant growth and consequently productivity of soils.

Specific Competence.

Demonstrate understanding of nutrient cycles.

Learning Activities

The following are the activities you can use to teach learners about soil fertility and nutrient cycles.

2.1.1 Analysing the nutrient cycles

Learning Materials: Textbooks, charts of nutrient cycles, worksheets, soil samples, compost manure...



Activity 1. Analysing the nutrient cycles

Learning materials: Notebooks

Procedure

- i. Divide learners into small groups of 3-4.
- ii. Ask learners to visit a school garden or nearby farming area.
- iii. Ask learners to observe the general appearance of growing crops or existing vegetation using their previous knowledge on soil fertility.

- iv. Ask learners to deduce their observations in relation to nutrients of growth.
- v. Ask learners to present their findings.

Content Tip:

- Emphasis should be on how the plants are benefiting from the nutrient cycles (nitrogen, carbon...).

2.1.1 Analysing the nutrient cycles



Activity 2: Scenario

A farmer in Kaputa district, Northern province experienced poor yields in crop production during the 2024/2025 farming season. Leaching was found to be one of the factors that led to low yields as observed by a local agricultural extension officer that visited his farm. The farmer was advised to do the following to improve the crop yields;

- ✓ Practising crop rotation
- ✓ Intercropping
- ✓ Use of organic manure

Procedure:

- i. Divide learners into small groups of 3-4
- ii. Ask learners to discuss the advice the farmer was given to improve crop yields.
- iii. Ask learners to present their findings.

Supporting activities

- i. Consulting digital and written media.

2.1.2 Illustrating the nitrogen and the carbon cycles

Learning Materials: Charts, markers,



Activity 1. Illustrating the nitrogen and the carbon cycles

Procedure:

- i. Divide learners into small groups of 3-4
- ii. Ask learners to draw a nitrogen cycle diagram showing (atmosphere, soil, plants, animals and bacteria). Learners should show the correct sequence of the following processes (nitrogen fixation, nitrification, assimilation, ammonification and denitrification).
- iii. Ask learners to discuss the importance of these processes in step (ii) above.
- iv. Ask learners to present their findings.



Content Tip

- This activity helps learners to develop practical skills, teamwork, observation and understanding of how nutrient cycles maintain soil fertility.

2.1.2 Illustrating the nitrogen and the carbon cycles

Learning Materials: Charts, markers,



Activity 2. Illustrating the Nitrogen and the Carbon cycles

Procedure:

- i) Divide learners into small groups of 4 – 5.
- ii) Ask learners to design models of the nutrient cycles (Nitrogen cycle and carbon cycle).
- iii) Ask learners to identify and label the main processes in each cycle.
- iv) Ask learners discuss how nutrients move between soil, plants and animals.
- v) Ask learners to present their findings

Supporting activities

- Viewing video clips on nutrient cycles.
- Consulting electronic or printed materials

2.1.3 Applying the nitrogen and carbon cycles in relation to soil fertility

Learning Materials: Dry leaves, straw, paper, green leaves, kitchen waste, manure



Activity 1. Applying the nitrogen and carbon cycles in relation to soil fertility

Procedure

- i. Divide learners into small groups of 3-4
- ii. Ask learners to create a compost heap using carbon sources (dry leaves, straw, paper), nitrogen sources (green leaves, kitchen waste, manure).
- iii. Ask learners to monitor rate of decomposition, temperature and moisture content.
- iv. Ask learners to explain how composting improves soil carbon and nitrogen content in relation to soil fertility.
- v. Ask learners to present their findings.

Supporting activities

- Viewing video clips on nutrient cycles.

- Consulting electronic or printed materials



Skills Developed

- **Communicating** – Learners working together to share their findings on nutrient cycles
- **Collaborating** – Learners tackling given tasks in groups
- **Analytically Thinking** – Analysing the transformation of various compounds into required nutrients.
- **Problem Solving** – Learners using knowledge of nutrient cycles to solve soil fertility problems
- **Environmental Sustainability** – Efficient utilisation of various nutrients
- **Creating and Innovating** – Learners making compost

Expected Standard

Understanding of nutrient cycles demonstrated accordingly.



Assessment

The Provincial Department of Agriculture has organized a provincial conference titled “Future of Soil: Maintaining Its Fertility for Future Productivity.” The conference aims to promote awareness of soil management and sustainable agricultural practices among communities, farmers, and learners.

As learners who have acquired knowledge about nutrient cycles, particularly the nitrogen cycle and carbon cycle, your school has been invited to participate in the conference. You are required to prepare key points that will help participants understand how soil nutrients are maintained and how human activities influence these natural processes.

Task 1

Prepare key points for your presentation addressing the following:

- i. Describe the term soil fertility;
- ii. Identify the essential nutrients that plants obtain from the soil;
- iii. Describe the nitrogen cycle and outline its major processes;
- iv. Explain the carbon cycle and discuss its significance in soil productivity and plant growth, and
- v. Explain how human activities can affect nutrient cycles and soil fertility.



Sub-Topic 2.2.1 Summary

In the nitrogen cycle, nitrogen moves between the soil, plants, animals and atmosphere, returning to the soil through decomposition and waste. In the carbon cycle, carbon moves between plants, animals, soil and the atmosphere through processes like photosynthesis, respiration and decomposition. Understanding these cycles helps farmers maintain fertile soil, grow healthy crops and manage land sustainability.

TOPIC 2.3.0. CLIMATE - SMART AGRICULTURE



Introduction

This topic focuses on climate change impact on agriculture and food security. The learners will explore effects of climate change in agriculture focusing on how crops and animal production is affected. Learners will further explore how agricultural activities contribute to climate change and discuss food security and its four dimensions.

Hook “Climate-Smart Agriculture helps farmers produce more food today while protecting tomorrow’s environment.”



General Competencies

- Analytical thinking
- Communication
- Collaboration
- Critical thinking
- Environmental sustainability
- Problem solving



Key Concepts/Procedures/Terms

Accessibility - How easy it is for people to obtain something

Adaptation- Adjusting farming methods to reduce the effects of climate change.

Availability-	State of being present or ready when needed
Climate change-	Long term changes in terms and weather patterns (rainfall, temperature and wind...).
Global warming -	Increase in average global temperatures over a long period of time.
Food Security-	Vital aspects of human life ensuring everyone has access to sufficient safe and nutritional food for active and health life.
Mitigation -	measures taken to reduce the impact of agriculture on climate such as reducing emissions from livestock.
Stability -	Not easily changing/failing apart.

Sub-Topic 2.3.1. Climate Change Impact on Agriculture and Food Security



Introduction

In this sub-topic, learners will explore the effects of climate change on agriculture and how agricultural activities contribute to climate change. In addition, learners will understand the relationship of climate change impact on agriculture and food security.

Specific Competence:

Demonstrate understanding of the relationship of climate change agriculture and food security.

Learning activities:

The following are the activities you can use to teach learners on climate change impact on agriculture and food security.

3.1.1 Exploring effects of climate change on agriculture (effects on crop and animal production).

Learning Materials: Pictures, posters, charts, maps, textbooks with information on climate change and farming.



Activity 1. Exploring effects of climate change on agriculture (effects on crops and animal production).

Procedure

- i. Divide learners into small group of 3-4
- ii. Ask learners to observe pictures, charts, maps related to climate change and farming.
- iii. Ask learners to identify climate change effects on observed materials provided.
- iv. Ask learners to report their findings.

Supporting Activities

- Field visit/school garden observation
- Case study analysis

3.1.2 Exploring how agricultural activities contribute to climate change

Learning Materials: Notebook



Activity 1. Exploring how agricultural activities contribute to climate change

Procedure

- Divide learners into small groups of 3-4
- Ask learners to visit a nearby farm or agricultural field to observe the following farming practices;
 - Burning of crop residues
 - Use of chemical fertilizers
 - Livestock keeping
 - Land clearing
 - Deforestation for farming
- Ask learners to record activities observed.
- Ask learners to present how the recorded activities contribute to climate change.



Content Tip

- Burning vegetation and keeping large number of livestock release greenhouse gases that contribute to climate change.
- Agriculture releases gases such as carbon dioxide, methane, and nitrous oxide, which contribute to Climate Change.

3.1.2 Exploring how agricultural activities contribute to climate change

Learning materials: Flip charts, markers, manilla papers



Activity 2: Debate topic

Hook “Modern agriculture contributes more to climate change than traditional agriculture.”

Procedure

- i. Divide learners into two groups.
- ii. Ask one group to support the statement and the other oppose it.
- iii. Ask each group to summaries its arguments with examples.
- iv. Ask learners to make posters of the main points raised in the debate from both sides.



Content Tip

- Teacher to consolidate arguments or points raised by the learners

Supporting Activities

- i. Viewing video clips of climate change impact by agricultural activities.
- ii. Participating in agricultural field days, shows and exhibitions

3.1.3 Discussing food security and its four dimensions (availability, accessibility, utilisation and stability)

Learning Materials: Textbooks, Notebook



Activity 1. Discussing food security and its four dimensions (availability, accessibility, utilisation and stability)

Procedure

- i. Divide learners into small groups of 3-4
- ii. Ask learners to visit community market and list common foods available.
- iii. Ask learners to create a community food basket.
- iv. Ask learners to discuss whether these foods are always available in homes.
- v. Ask learners to suggest where these foods come from (farm, market, imports?)
- vi. Ask learners to present their findings.



Content Tip

- This activity focuses on dimension ‘availability’
- Food availability depends on production, storage, and supply.

3.1.3 Discussing food security and its four dimensions (availability, accessibility, utilisation and stability)



Activity 2: Study tour

Procedure

- i. Learners to visit a nearby market or shop.
- ii. Ask learners to interview shopkeepers, marketeers or observe whether the commodities are available or not
- iii. Ask learners to record the information observed.
- iv. Ask learners to present their findings.



Content Tip

- This activity focuses on dimension ‘accessibility
- Accessibility – people must have money, resources, and access to obtain food.

3.1.3 Discussing food security and its four dimensions (availability, accessibility, utilisation and stability)

Learning materials: Charts, diagrams



Activity 3: Discussing food security and its four dimensions (availability, accessibility, utilisation and stability)

Procedure

- i. Put learners into small groups of 4-5
- ii. Give learners different food samples
- iii. Ask learners to categorize the food samples into carbohydrates, proteins, fats, vitamins, minerals and dietary fibre
- iv. Ask learners to make a simple chart of how each nutrient is used in the body
- v. Let learners discuss the type of nutrient in the food samples, how those nutrients are utilized in the body and what happens if the body cannot utilize the food properly.
- vi. Ask learners to make their presentations to the class.



Content Tip

- This activity focuses on dimension ‘Utilisation’
- Utilisation - the body must properly use the nutrients in food through good diet, sanitation, and health.

3.1.3 Discussing food security and its four dimensions (availability, accessibility, utilisation and stability)



Activity 4: Scenario 1:

Most small-scale farmers in Zambia depends on rain fed agriculture for food and income. When the rainy season failed to arrive, the land became dry, rivers and wells began to shrink, and crops withered before they could grow. As a result, the harvest failed and many families faced food shortages.

- i. Put learners into small groups 4-5.
- ii. Ask learners to discuss the scenario presented above.
- iii. Ask learners to make presentation on solutions to the problems in the scenario above.



Content Tip

- This activity focuses on dimension ‘Stability’
- Techniques in proper food storage, improved water harvesting, irrigation, relief food

Supporting Activities

- i. Visiting local farmers
- ii. Viewing videos
- iii. Consulting printing and electronic media



Skills Developed

- **Communicating** – Learners working in groups and engaging community members
- **Collaborating** – Working in groups to carryout assigned activities
- **Analytically Thinking** – Analysing relationship between climate change and food security

- **Problem Solving** – Learners finding ways of securing food in times of weather changes
- **Environmental Sustainability** - Learners using mitigative interventions
- **Creating and Innovating** – Learners combining various dimensions to ensure sustainable food security

Expected Standard

Understanding of relationship of climate change, agriculture and food security demonstrated accordingly.



Assessment

A small-scale farmer in a rural community has planted maize at the beginning of the growing season. However, the area has recently experienced low and unreliable rainfall, causing the maize seedlings to begin drying up and showing signs of moisture stress. The farmer is concerned that continued dry conditions may lead to poor crop growth and reduced yields. As a learner who has studied climate-smart agricultural practices, you are asked to advise the farmer on possible solutions to reduce the negative impact of drought conditions on the crop.

- Suggest three climate-smart agricultural practices the farmer can adopt to reduce the impact of low rainfall on the maize crop and briefly explain how each practice helps the crop survive under dry conditions.
- During a recent flood season, a community lost a significant number of goats due to rising waters and poor grazing conditions. This event has raised concerns about the impact on household livelihoods and food security. As a learner who has studied climate change, livestock management, and food security, you are asked to analyze the situation and provide guidance to the community.
 - i. Discuss two effects of the loss of goats on household food security;
 - ii. Explain how high temperatures can affect livestock, and
 - iii. Discuss reason why climate change can lead to food insecurity.



Sub-Topic 2.3.1 Summary

Climate - Smart Agriculture is an approach that increases farm productivity. It builds resilience to climate change and reduces emissions. Climate change affects crops and livestock through lower yields in crops and reduced growth and fertility in livestock. Climate - Smart Agriculture addresses these effects through, adaption, mitigation and productivity. It helps sustain crop and animal production, improves resilience to climate variability and supports food security while minimizing environmental impact. Climate-Smart agriculture helps farmers face these challenges while protecting the environment.

TOPIC 2.4.0 FARM STRUCTURES AND MACHINERY



Introduction

Farm structures and machinery are important components of agriculture production. Farm structures and buildings and facilities on the farm used for housing animals, string crops and protecting equipment. Farm machinery includes tools and machines used to carry out farm operations such as ploughing, planting and harvesting. The use of proper farm structures and machinery helps to reduce labour, save time and increase efficiency in farming. Understanding this topic equips learners with basic skills needed for effective farm management.



General Competences

- Analytical thinking,
- Communication,
- Creativity and innovation,
- Collaboration,
- Critical thinking,
- Environmental sustainability and
- Problem solving.

Hook: *“How can one farmer produce more food with less effort?”*



Key Concepts / Processes / Terms

- Farm structures** - Buildings on the farm used to support agricultural activities.
- Farm Machinery** - Tools and machines to carryout farm work.
- Hand tools** - Simple farm tools operated by hand.
- Implements** - Equipment used with animals or machines to perform farm tasks.
- Maintenance** - Care and repair of farm tools and machinery.
- Efficiency**- Doing farm work with less time and effort.

Sub-Topic 2.4.1 Farm Buildings



Introduction

Farm buildings are structures constructed on the farm to support agricultural activities. They are used for housing livestock, storing crops and equipment, and protecting farm inputs. Proper farm buildings improve efficiency, safety and organisation in farm operations.

Specific Competence

Construct farm structures

Learning activities

The following are the activities you can use to teach learners about farm buildings

4.1.1 Exploring a suitable site for farm buildings (*security, direction of prevailing wind, drainage, accessibility, relationship between structures...*)

Learning materials: Field notebook



Activity 1: Exploring a suitable site for farm buildings (*security, direction of prevailing wind, drainage, accessibility, relationship between structures...*)

Procedure

- i. Divide learners into small groups of 3-4
- ii. Ask learners to visit a nearby farm or open land.
- iii. Ask learners to observe possible sites where farm buildings could be constructed.
- iv. Ask learners to record their observations.
- v. Ask learners to present their findings in class.



Content Tip:

- Learners identify appropriate locations for different farm buildings.
- Factors to consider for a suitable site for farm buildings should include; Security of the area, direction of the prevailing wind, orientation of the building (wind, sunlight, roads, surrounding environment), drainage of the land, accessibility by road or path, movement

4.1.2 Preparing a basic layout and plans for farm buildings.

Learning materials: Charts, Plain papers



Activity: Preparing a basic layout and plans for farm buildings.

Procedure

- i. Put learners into small groups of 3-4
- ii. Ask learners to design and draw a simple farm layout showing the positions of Farm house, Animal houses, storage facilities, roads...)
- iii. Ask learners to make presentations on their drawings



Content tip:

- Learners must arrange buildings considering: Accessibility, security, proper spacing, orientation

4.1.3 Building farm structures using local materials (*roads, farmhouse, fences...*)

Learning materials: Wood or poles, grass or thatch, stones, clay soil, bamboo



Activity 1: Building farm structures using local materials (*roads, farmhouse, fences...*)

Procedure

- i. Put learners into small groups of 3-4
- ii. Ask learners to choose the type of farm building to construct using local materials

- iii. Ask learners to walk around the school or community and identify local materials available that can be used to build farm structures chosen.
- iv. Ask learners to construct the selected farm structure using selected local materials



Content tip

- Local materials may include; Wood or poles, grass or thatch, stones, clay soil, bamboo...



Skills Developed

- **Communicating** – Learners working in groups and engaging community members.
- **Collaborating** – Working in groups to carryout assigned activities.
- **Analytically Thinking** – Comparing of operations for animal and tractor drawn implements.
- **Problem Solving** – Learners finding ways of maintaining animal and tractor drawn implements.
- **Environmental Sustainability** - Use of local materials which are degradable for building farm structures.
- **Creating and Innovating** – Learners building farm structures using local materials.

Expected Standard

Farm structures constructed correctly.



Assessment

- 1) List four environmental factors to consider when choosing a site for farm structures.

- 2) Sketch a simple farm layout and indicate the most suitable site for livestock shelter.
- 3) Explain why site selection is important before constructing farm buildings.
- 4) A proposed site is near a low-lying area prone to flooding. Suggest an alternative location to explain your reasoning.
- 5) Explain how proper site selection contributes to durability, cost savings and efficient farm management.



Sub-Topic 2.4.1. Summary

Farm buildings support crop and livestock production. A suitable site should have good drainage, accessibility, safety from hazards and proximity to resources. Exploring the site helps plan durable, functional and efficient structures, improving farm productivity and management.

Subtopic 2.4.2. Animal and Tractor Drawn Implements



Introduction

In agriculture, implements are tools or equipment used to carry out farm tasks more efficiently. In this sub-topic, learners will study animal and tractor drawn implements which are essential for land preparation, and other farm operations. Animal drawn implements are attached to draught animals like oxen and are used for tasks such as ploughing, harrowing and cultivating small farms. Tractor drawn implements are attached to a tractor and include tools like ploughs, harrows, cultivators, and planters. These increase efficiency and reduce manual labour in larger farms. Studying these implements helps learners identify, describe, use and maintain them correctly, improving productivity and reducing effort in agricultural activities.

Specific Competence

Operate animal and tractor drawn implements

Learning Activities

The following are the activities you can use to teach learners about animal and tractor drawn implements;

1.2.1 Comparing animal and tractor drawn implements

Learning materials: Pictures of animal and tractor drawn implements or animal and tractor drawn implements



Activity 1: Comparing animal and tractor drawn implements

Procedure

- i. Put learners into small groups of 4-5.
- ii. Ask learners to observe materials provided.
- iii. Ask learners to identify animal and tractor drawn implements from the materials provided with reasons.
- iv. Ask learners to present their findings.

1.2.2 Preparing animal and tractor drawn implements for appropriate use (mouldboard, disc plough, ridger, planter, ripper, cultivator, boom sprayer, harrow, subsoiler...)

Learning materials: Mouldboard, disc plough, ridger, planter, ripper, cultivator, boom sprayer, harrow, subsoiler



Activity 1: Preparing animal and tractor drawn implements for appropriate use (mouldboard, disc plough, ridger, planter, ripper, cultivator, boom sprayer, harrow, subsoiler...)

Learning Materials; Field notebook, Charts, Mannila paper

Activity 1. Field Tour

Procedure

- i. Ask learners to visit a farm mechanization centre or any place where they can find farm implements.
- ii. Divide learners into small groups of 4-5.
- iii. Ask learners to prepare animal and tractor drawn implement for appropriate use.
- iv. Ask learners to make a presentation on how they prepared the implements.



Content Tip:

- Teacher should demonstrate proper handling of animal and tractor drawn implements and explain reasons for preparing these implements.

- Learners inspect an implement and check for loose bolts and nuts, worn or damaged parts, lubrication points, tyres or discs condition.

1.2.3 Using animal and tractor drawn implements

Learning materials: mouldboard plough, disc plough, ridger, planter, cultivator, boom sprayer



Activity 1: Using animal and tractor drawn implements

Procedure

- Put learners into small groups of 4-5
- Ask learners to operate animal and tractor drawn implements.
- Ask learners in their various groups of operations to take notes for discussion later.
- Ask each group to present how they operated different animal and tractor drawn implements.



Content Tip

- The teacher should demonstrate correct operational procedures of the farm implements at school or visited site.

1.2.4 Maintaining animal and tractor drawn implements appropriately

Learning materials: Tractor, oil, grease, cleaning equipment, water, brushes, and cloths



Activity 1: Maintaining animal and tractor drawn implements appropriately

Procedure

- i. Take learners to a site with animal and tractor drawn implements.
- ii. Let learners in groups of 4-5 participate in all or any of following maintenance operations;
 - cleaning implements using water, brushes, and cloths,
 - removing soil, plant residues, and chemicals from implements.
 - drying the implements to prevent rusting.
 - applying oil or grease to moving parts such as bearings, joints, and wheels.
 - tightening loose fittings.
 - checking for blockages or leakages.
- iii. In their groups, let learners take notes of what they have observed for presentation to the class



Skills Developed

- **Communicating** – Working in groups to prepare and maintain animal and tractor drawn implements
- **Collaborating** – Learners work together when using animal and tractor drawn implements.
- **Analytically Thinking** – Learners are able understand how animal and tractor drawn implements operate
- **Problem Solving** – Learners are able to repair animal and tractor drawn implements
- **Environmental Sustainability** – Proper preparation, efficient utilisation and maintenance contribute to environmental sustainability.
- **Creating and Innovating** – Learners will innovate by finding the best methods of preparing and maintaining animal and tractor drawn implements.

Expected Standard

Animal and tractor drawn implements operated accordingly.



Assessment

In a rural farming community, farmers rely on animal-drawn and tractor-drawn implements to prepare land for planting crops. Proper use of these implements is essential to improve soil fertility, ensure crop success, and maintain safety during farm operations.

As a learner with knowledge of farm implements and mechanized farming practices, you are required to demonstrate understanding of both animal-drawn and tractor-drawn equipment.

- i. Name three animal-drawn implements used for soil preparation and state the function of each.
- ii. Explain the steps to safely attach a tractor-drawn implement before use.

In a rural agricultural community, farmers use both animal-drawn and tractor-drawn implements to prepare land for crop production. Proper maintenance and safe operation of these tools are crucial to ensure efficiency, longevity, and safety on the farm.

As a learner with knowledge of farm implements, their maintenance, and safe operation, you are required to demonstrate your understanding of these practices.

- i. Describe one common maintenance practice for a plough.
- ii. Explain the safety precautions that should be taken when operating a harrow with animals.
- iii. Compare the advantages of tractor-drawn implements over animal-drawn implements.



Sub-Topic 2.4.1. Summary

Implements are tools used in farming to prepare soil, plant, cultivate and transport crops. Animal-drawn implements include ploughs, harrows, seed drills and carts which are powered by animals. Tractor-drawn implements include ploughs, cultivators, harrows, planters, sprayers and trailers. These perform tasks like ploughing, harrowing, seeding, cultivating and transporting produce. Safe

handling, proper attachment and correct operation are essential to prevent accidents. Regular maintenance like cleaning, lubrication and minor repairs ensures efficiency and longevity. This topic equips learners with practical and theoretical knowledge on farm structures and farm machinery, emphasizing the ability to plan, construct, operate, and maintain farm buildings and implements for effective agricultural production.

TERM 2

TOPIC 2.0 CROP PRODUCTION



Introduction

This topic focuses on crop production. Learners will develop an understanding of both the external and internal morphology of plants, as well as the principles of field crop production. By the end of the topic, learners would have produced a crop and acquired practical knowledge and skills in field crop production, beginning with plant morphology, land preparation and extending to innovative methods of increasing agricultural productivity.



General Competences

- Analytical thinking,
- Communication,
- Creativity and innovation,
- Collaboration,
- Critical thinking,
- Environmental sustainability and
- Problem solving.

Hook: *“What would happen if farmers stopped growing crops for just one year?”*



Key Concepts / Processes / Terms

Disease -	an abnormal condition affecting plant growth.
Dormancy -	a resting period when a seed does not germinate even under favourable conditions.
External morphology -	visible plant parts such as roots, stems, leaves, flowers, fruits, and seeds.
Germination -	the process by which a seed develops into a seedling.
Harrowing -	breaking soil clods and leveling the field.
Irrigation -	artificial application of water to crops.
Internal morphology -	internal structure of plants such as tissues and cells.
Mulching-	The practice of covering soil around plants with organic or synthetic material to retain moisture, suppress weeds, and improve soil fertility.
Pest-	an organism that damages crops, reducing yield or quality.
Pathogens-	microorganisms such as fungi, bacteria, or viruses that cause disease in plants.
Seed -	a fertilised ovule capable of developing into a new plant.
Seed viability -	the ability of a seed to germinate and grow.
Storage -	keeping harvested crops safely to prevent losses.
Threshing -	separating grains from stalks or husks.
Winnowing -	separating grain from chaff using wind or air.
Weeding -	the removal of unwanted plants (weeds) that compete with crops for nutrients, water, and sunlight.

Sub-Topic 2.5.1 External and Internal Morphology of Plants



Introduction

In this sub-topic, learners will classify different types of crops according to their life cycles. They will examine and analyse the external and internal morphology of plants and relate the structure of various plant parts to their functions. Learners will also compare different types of plant propagation, identify the suitable conditions required for seed germination and practice various methods of vegetative propagation.

Specific Competence

Demonstrate understanding of external and internal morphology of plants

Learning Activities

The following are some of the activities you can use to teach the learners about the external and internal morphology of plants.

5.1.1 Classifying different types of crops based on their life cycle (annual, biennial and perennial)

Learning Materials: Cabbage, onion, tomatoes, maize, groundnuts, cassava, lemons, oranges, pawpaw, mango, charts.



Activity 1. Classifying different types of crops based on their life cycle (annual, biennial and perennial)

Procedure

- i. Put learners into small working groups of 4 -5.
- ii. Ask learners to research and prepare a presentation on the meaning of annual, biennial and perennial crops with examples.
- iii. Ask learners to report their findings.



Discussion

1. A farmer in your community wants to grow different crops but does not understand how long different crops take to complete their life cycles.
 - a) Explain the meaning of the following types of crops:
 - i. Annual crops
 - ii. Biennial crops
 - iii. Perennial crops



Content Tip

The following crops are classified based on life cycles;

- Annual crops: Maize, beans, rice,
- Biennial crops: Cabbage, carrot, onions, beetroot
- Perennial crops: Mango, banana, sugarcane, cassava

Supporting Activities

- i. Consult digital resources and print media

5.1.1 Classifying different types of crops based on their life cycle (annual, biennial and perennial)

Learning materials: Notebook



Activity 2. Study tour

Procedure

- v. Put learners into small working groups of 4 - 5.
- vi. Ask learners to visit a farm, school garden or vegetable market to identify different types of crops commonly grown locally.
- vii. Ask learners to classify the crops identified during the visit based on its life cycles (annual, biannual and perennial crops).
- viii. Ask learners to report their findings through presentations to the class.

Supporting Activities

- Consulting print media and digital resources
- Viewing video clips

5.1.2 Analysing the external and internal morphology of plants (roots, stems, leaves and flowers)

Learning Materials: Field notebook, pencil, hand lens or magnifying glass, plant identification guide, different plant species.



Activity 1. Analysing the external and internal morphology of plants (roots, stems, leaves and flowers)

Procedure

- i. Put learners into small groups of 4 – 5

- ii. Ask learners to visit a garden, field or woodland.
- iii. Ask learners to pick three (3) plant species.
- iv. Ask learners to examine and draw the different basic parts of plants
- v. Ask learners to discuss the functions of each part of the plant examined.
- vi. Ask each group to present their findings.



Discussion

A student observes a plant with broad flat leaves, a green stem and fibrous roots.

- i. Identify the three external parts of the plant.
- ii. State one function of each part.

5.1.2 Analysing the external and internal morphology of plants (roots, stems, leaves and flowers)

Learning Materials: Surgical blade or scalpel, hand lens or microscope, magnifying glass, slides and covers, water, charts or diagrams of plant internal structures, worksheet, observation table, notebook, pencil, plant identification guide.



Activity 2. Root internal structure

Procedure

- i. Put learners into small groups of 4 – 5
- ii. Ask learners to visit a garden, field or woodland. Learners should select three different types of plants.

- iii. Ask learners to cut a thin cross section of a root.
- iv. Ask learners to observe using a hand lens or magnifying glass and draw the internal structure.
- v. Ask learners to identify and discuss functions of the internal parts observed.
- vi. Ask each group to present their findings.



Content Tip

Learners should identify the following structures (cortex, epidermis, xylem, phloem).

5.1.2 Analysing the external and internal morphology of plants (roots, stems, leaves and flowers)

Learning Materials: Surgical blade or scalpel, hand lens or microscope, plant stems, magnifying glass, slides and covers, water, charts or diagrams of plant internal structures, worksheet, observation table, notebook, pencil, plant identification guide.



Activity 3. Stem internal structure

Procedure

- i. Put learners into small groups of 4 – 5.
- ii. Ask learners to visit a garden, field or woodland. Learners to pick three different types of plants.
- iii. Ask learners to cut a thin cross section of a stem.
- iv. Ask learners to observe using a hand lens or magnifying glass and draw the observed parts.
- v. Ask learners to observe using a hand lens or magnifying glass and draw the internal structure.

- vi. Ask learners to identify and discuss functions of the internal parts observed.
- vii. Ask each group to present their findings.



Content Tip

- Learners to identify the following structures (cortex, epidermis, xylem, phloem, pith).

5.1.2 Analysing the external and internal morphology of plants (roots, stems, leaves and flowers)

Learning Materials: Surgical blade or scalpel, hand lens or microscope, magnifying glass, slides and covers, water, charts or diagrams of plant internal structures, worksheet, observation table, notebook, pencil, plant identification guide.



Activity 4. Leaf internal structure

Procedure

- i. Put learners into small groups of 4 – 5.
- ii. Ask learners to visit a garden, field or woodland. Learners should pick three different leaves of plants.
- iii. Ask learners to prepare slides of a leaf.
- iv. Ask learners to observe and examine the internal structure of a leaf using a hand lens or magnifying glass
- v. Ask learners to draw a labelled diagram of the internal structure of a leaf.
- vi. Ask learners to discuss the functions of the parts observed.

- vii. Ask learners to present their findings.



Content Tip

Learners to identify the following structures (upper epidermis, palisade mesophyll, spongy mesophyll, vascular bundles).

5.1.2 Analysing the external and internal morphology of plants (roots, stems, leaves and flowers)

Learning Materials: Surgical blade or scalpel, hand lens or microscope, magnifying glass, slides and covers, water, charts or diagrams of plant internal structures, worksheet, observation table, notebook, pencil, plant identification guide, flowers (e.g., hibiscus or lily)



Activity 5: Analysing the external and internal morphology of plants (roots, stems, leaves and flowers)

Procedure

- i. Put learners into small groups of 4 – 5.
- ii. Ask learners to visit a garden, field or woodland. Learners to pick different flowers from plants.
- iii. Ask learners to dissect a flower.
- iv. Ask learners to observe using a hand lens or magnifying glass
- v. Ask learners to draw a labelled diagram of the structure of a flower.
- vi. Ask learners to discuss the functions of the parts observed.
- vii. Ask learners to present their findings.



Content Tip

- Parts of the flower: Sepals, petals, stigma, style, ovary, anthers, filaments

Supporting Activities

- i. Consulting printing and electronic media

5.1.3 Drawing and relating the structure of the parts of the plant to their functions (for a flower include pollination and fertilization)

Learning materials: Charts, different flowers from the plants



Activity 1: 5.1.3 Drawing and relating the structure of the parts of the plant to their functions (for a flower include pollination and fertilization)

Procedure

- i. Put learners in small groups of 4-5
- ii. Ask learners to visit a garden, field or woodland. Learners to pick different flowers from plants (e.g., hibiscus or lily)
- iii. Ask learners to dissect them.
- iv. Ask learners to observe and draw a structure of a flower
- v. Ask learners to identify and label different parts of the flower
- vi. Ask learners to discuss the functions of the parts identified.
- vii. Ask each group to present their findings.

5.1.4 Comparing types of propagation (seed and vegetative)

Learning Materials: Seeds (maize, beans, groundnuts), vegetative planting material (cassava stem, sweet potato vine, potato tuber, onion bulb), markers, flip charts



Activity 1: Comparing types of propagation (seed and vegetative)

Procedure:

- i. Divide the learners into small groups of 3-5.
- ii. Ask learners to select samples of seeds and vegetative planting materials provided.
- iii. Ask the learners to observe the materials carefully and identify their differences.
- iv. Ask learners to create a comparison table with two columns; seed propagation and vegetative propagation
- v. Ask learners to compare aspects such as; planting material used, how the plant grows, examples of crops, it's advantages and disadvantages
- vi. Ask each group to present their findings



Content Tip

- The teacher should introduce the two types of propagation and then present a summary of the points after all the groups have presented their observations.

5.1.5 Relating the main parts of seed germination (plumule, radicle, seed testa, micropyle, cotyledon)

Learning Materials: Seeds (bean or maize), water, paper towels or cotton wool, small containers or plastics, magnifying glass, diagram of seed



Activity 1: Relating the main parts of seed germination (plumule, radicle, seed testa, micropyle, cotyledon)

Procedure

- i. Divide learners into small groups of 3-5
- ii. Ask each group to select a soaked seed (maize, bean) or seeds that have started germinating.
- iii. Ask learners to carefully split the seed into two halves.
- iv. Ask learners to examine and observe the internal parts of a selected seed.
- v. Ask learners to identify the parts of the germinating seed
- vi. Ask learners to draw the seed they observed and label the main parts.
- vii. Ask learners to briefly explain their labelled diagram to the class.



Content Tip

- Parts of a seed include; plumule, radicle, seed testa, micropyle, cotyledon).
- Learners should point out which part becomes the root, shoot and which part provides food for the seedling.

5.1.6 Identifying suitable conditions for seed germination (Temperature, oxygen, moisture, seed viability)

Learning Materials: Bean or maize seed, cotton wool or paper towel, four small containers or jars, water, refrigerator or cool place.



Activity 1: Identifying suitable conditions for seed germination (Temperature, oxygen, moisture, seed viability)

Procedure

- i. Divide learners into small groups of 3-5
- ii. Let each group prepare a different germination condition.

Group	Condition Tested	Set up
1	Moisture	Seeds placed in dry cotton wool (no water)
2	Oxygen	Seeds placed in cotton wool fully submerged in water (little oxygen)
3	Temperature	Seeds placed in moist cotton wool but kept in a cold place
4	Seed Viability	Use old or damaged seed in moist cotton at room temperature

- iii. Leave another set of seeds in moist cotton wool at room temperature as the control
- iv. Let learners observe for 3 – 5 days
- v. Each group should observe if germination has occurred or not and make a record
- vi. Each group should make a presentation on what they have observed

Supporting Activities

- i. Viewing video clips or pictures.
- ii. Consulting other digital or print media

5.1.6 Identifying suitable conditions for seed germination (temperature, oxygen, moisture, seed viability)

Learning materials: Seeds in small containers or Petri dishes



Activity 1: Identifying suitable conditions for seed germination (temperature, oxygen, moisture, seed viability)

Procedure

- i. Divide learners into small groups of 3-4.
- ii. Ask learners to set up seeds in small containers or Petri dishes with varying moisture levels: dry, slightly moist, and waterlogged.
- iii. Ask learners to observe germination over several days, focusing on radicle and plumule emergence.
- iv. Ask learners to record the number of seeds germinated under each condition.
- v. Ask learners to present their findings.



Content Tip

- Learners identify that moderate moisture supports optimal germination.
- Explains the role of water in activating metabolic processes in the seed.

Supporting Activities

- i. Viewing video clips or pictures.
- ii. Consulting other digital or print media

5.1.7 Practicing different types of vegetative propagation (*budding, grafting, layering, cutting, rhizomes, bulbs, corms...*) for suitable crops

Learning materials: Vegetative propagation, Knife or razor blade, Secateurs, grafting tape or polythene strips, small pots or nursery bags, soil or potting mixture, watering can, Plant materials (e.g., cassava stems, rose stems, ginger rhizomes, onion bulbs, taro corms)



Activity 1: Practicing different types of vegetative propagation (*budding, grafting, layering, cutting, rhizomes, bulbs, corms...*) for suitable crops

Procedure:

- i. Divide learners into small groups of 4-5
- ii. Assign each group one vegetative propagation method.
- iii. Provide plant materials and tools for each group.
- iv. Each group performs the method as follows:
 - Cutting: Prepare stem cuttings and plant them in soil.
 - Layering: Bend a branch to the ground, cover part with soil, leaving the tip exposed.

- Budding: Insert a bud from one plant into the stem of another plant.
 - Grafting: Join a scion and rootstock and tie them with grafting tape.
 - Rhizomes: Plant pieces of rhizomes such as ginger in soil.
 - Bulbs: Plant bulbs such as onions with the tip facing upward.
 - Corms: Plant corms such as taro in prepared soil.
- v. Ask learners to label their plants and water them.
- vi. Ask each group to present and explain the method they practiced to the class.

Supporting Activities

- i. Viewing video clips or pictures.
- ii. Consulting other digital or print media



Skills Developed

- **Communicating** – Learners presenting their work to the class
- **Collaborating** – Learners working together in groups during assigned tasks
- **Analytically Thinking** – Analysing the external and internal morphology of plants in relation to their functions.
- **Problem Solving** – Producing plants using vegetative propagation methods in place of expensive seed propagation method.
- **Environmental Sustainability** – Sustaining the environment through use of vegetative propagation

Expected Standard

Understanding of external and internal morphology of plants demonstrated appropriately



Assessment

Task 1

A school agriculture club has started a garden demonstration to help learners understand plant growth and crop production. The club intends to grow different crops, observe plant structures, and practice plant propagation techniques to improve crop production in the community. As a learner with knowledge of crop classification, plant morphology, and plant propagation, you have been asked to guide other club members on how plants grow and reproduce.

1. Classify crops according to their life cycles and give examples of each type;
2. Describe the external and internal morphology of plants, highlighting the main parts and their roles in plant growth;
3. Draw and label a plant and explain how the structure of each part relates to its function;
4. Identify suitable conditions required for plant growth and development, and
5. Explain and demonstrate different methods of vegetative propagation that can be used in the demonstration garden.



Sub-Topic 2.5.1 Summary

In this sub-topic, learners developed understanding of plant growth, structure, and reproduction in crop production. It emphasizes the ability to classify crops, analyze plant structures, relate plant parts to their functions, identify suitable growth conditions, and practice vegetative propagation techniques used in agriculture. This sub-topic helped learners to build practical agricultural knowledge and scientific understanding of plant growth and reproduction, which are essential for improving crop production and sustainable farming practices.

Sub-Topic 2.5.2 Field Crop Production



Introduction

This subtopic focuses on field crop production with particular emphasis on the cultivation of cereal crops. Learners will explore the types of cereal crops commonly grown locally and gain an understanding of how to select suitable sites for successful cereal crop production. The topic also introduces key technical aspects such as determining plant population, calculating seed germination percentage and estimating the required inputs per hectare. In addition, learners will examine essential crop management practices throughout the production cycle, from planting to harvesting. Finally, learners will explore innovative approach and improved practices aimed at increasing productivity and enhancing the sustainability of cereal crop production.

Specific Competence

Grow cereal crops

Learning Activities

The following are some of the activities you can use to teach the learners about growing cereal crops.

5.2.1 Exploring cereal crops locally grown (*maize, sorghum, wheat, rice, millet, barley...*)

Learning Materials: Maize, sorghum, wheat, rice, millet, barley



Activity 1: Exploring cereal crops locally grown (*maize, sorghum, wheat, rice, millet, barley...*)

Procedure

- i. Put learners into small groups of 4 – 5.

- ii. Ask learners to visit a farm, school garden or a market to identify different types of cereal crops commonly grown locally.
- iii. Ask learners to list all cereal crops identified and discuss their nutritional and economic value.
- iv. Ask each group of learners to make presentations to the class.

Supporting Activities

- i. Viewing video clips
- ii. Consulting electronic or printed media

5.2.2 Selecting a site suitable for cereal crop production

Learning material: Notebook



Activity 1: Selecting a site suitable for cereal crop production

Procedure

- i. Put learners into small groups of 4 – 5.
- ii. Ask learners to visit a school garden.
- iii. Ask learners to look at the characteristics of a site suitable for cereal crop production in the school garden.
- iv. Ask learners to demarcate a portion where they are going to grow cereal crops in groups.
- v. Ask learners to explain to the class how they arrived at the choices they made.
- vi. Ask learners to record the activity of the day.
- vii. Ask each group of learners to present their findings.

5.2.3 Designing a simple crop rotation plan including a cereal and legume crop (*cereal-legume-tuber and any other crops*)

Learning materials: Flip charts, manilla paper, cardboards, markers



Activity 1: Designing a simple crop rotation plan including a cereal and legume crop (*cereal-legume-tuber and any other crops*)

Procedure

- i. Put learners into small groups of 4 -5.
- ii. Ask learners to design a simple crop rotation model including a cereal and a legume.
- iii. Ask learners to present and explain their designs.
- iv. Ask learners to select the best crop rotation design and implement it in the field.
- v. Ask learners to note the best design in their dairies.



Content Tip

- Simple crop rotation model including a cereal and a legume in the rotation.

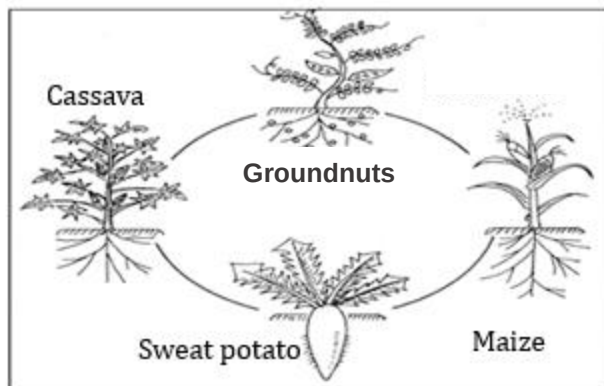


Figure 1.0

Supporting Activities

- i. Video clip of designing simple crop rotation.
- ii. A talk/visit on a local vegetable growing farmer

5.2.4 Determining plant population, seed germination and inputs required per hectare (*fertilizer, quantity of seeds, pesticides*)

Learning Materials: Measuring tape, Pegs or sticks, string, notebook and pen, calculator



Activity 1: Determination plant population

Procedure:

- i. Select a representative area in the field.

- ii. Use a measuring tape to mark a sample plot (e.g., 1 m × 1 m or 2 m × 2 m).
- iii. Place pegs at the corners and tie string to clearly mark the plot.
- iv. Count all the plants growing within the marked plot.
- v. Record the number of plants.
- vi. Repeat the counting in 3–5 different plots within the field to improve accuracy.
- vii. Calculate the average number of plants per plot.
- viii. Convert the result to plants per hectare using the formula below.

Formula:

$$\text{Plant population per hectare} = \frac{\text{Average number of plants in sample plot} \times 1000}{\text{Area of Sample Plot}}$$

Example:

Plot size = 2 m × 2 m = 4 m²

Average plants counted = 12 plants

$$\text{Plant population per hectare} = \frac{12 \text{ plants} \times 1000 \text{m}^2}{4 \text{m}^2} = 3000 \text{ plants per hectare}$$

5.2.4 Determining plant population, seed germination and inputs required per hectare (fertilizer, quantity of seeds, pesticides)

Learning Materials: Needed: Seeds (e.g., beans, maize, or ground nuts), petri dishes or small containers, cotton wool or paper towels, water, marker/labels, notebook and pen



Activity 2: Determining seed germination %

Procedure:

- i. Divide learners into small groups of 3–5 students.
- ii. Give each group 10–20 seeds.
- iii. Place moist cotton wool or paper towel in a container or petri dish.
- iv. Arrange the seeds evenly on the moist surface.
- v. Label each container with the group name and date.
- vi. Keep the containers in a warm place and ensure the cotton remains moist.
- vii. After 3–7 days, learners count the number of seeds that have germinated (seeds showing a radicle).

Calculation:

Learners calculate germination percentage using the formula:

$$\text{Germination \%} = \frac{\text{Seeds that have germinated}}{\text{Seeds planted}} \times 100$$

Example:

Seeds planted = 20

Seeds germinated = 16

$$\text{Germination \%} = \frac{16}{20} \times 100 = 80\% \text{ germination}$$

5.2.4 Determining plant population, seed germination and inputs required per hectare (*fertilizer, quantity of seeds, pesticides*)

Learning Materials: Calculator, Notebook and pen, Measuring tape, seed and fertilizer rates, Chalkboard/flip chart



Activity 3: Calculating inputs per hectare

Scenario

Given that the following inputs are required in the amounts stated below

- Maize seed rate: 25 kg per hectare
- Fertilizer rate: 200 kg per hectare

Group Activity

Divide learners into small groups and give them questions such as:

Example 1:

A farmer has 2 hectares of land.

How much maize seed will be needed if the rate is 25 kg/ha?

Calculation:

Maize seed needed per hectare = 25 kg × 2 ha = 50 kg of seed

Example 2:

A farmer wants to apply 200 kg of fertilizer per hectare on 3 hectares.

Calculation:

Fertilizer required for 3 hectares = 200 kg × 3 = 600 kg fertilizer

Field Measurement Extension (Optional Practical)

Take learners outside and measure a 10 m × 10 m plot (100m²).

Ask learners to calculate how much input would be needed for that plot based on per hectare rates.

Example:

Fertilizer rate = 100 kg/ha

Since 1 ha = 10,000 m²

For 100 m² plot

$$\text{Fertilizer required for 100 m}^2\text{plot} = \frac{100 \text{ m}^2}{10000\text{m}^2} \times 100 \text{ kg} = 1 \text{ kg}$$



Content Tip

- Teacher should explain that farmers must know how much seed, fertilizer, or chemicals are needed for one hectare (10,000 m²).

5.2.5 Managing maize and any other locally grown cereal crop from planting to harvesting (*field preparation, sowing/planting, spacing and plant population, planting depth, irrigation, weeding, fertilizer application, pest and disease control, and harvesting*).

Learning Materials Needed: Hoes or hand hoes, Pegs and string, Measuring tape, Slashers or cutlasses, Rakes, Field or garden plot



Activity- Managing maize and any other locally grown cereal crop from planting to harvesting

Activity 1: Field preparation

Procedure:

- i. Introduction (Class Discussion)
- ii. The teacher explains the importance of proper field preparation for maize production (removal of weeds, improving soil structure, and preparing land for planting).
- iii. Group Formation
- iv. Divide learners into small groups of 4–6.



Activity 2: Clearing the Field

- i. Each group clears a small section of the field by removing weeds, grasses, and crop residues using slashers or hoes.
- ii. Stumping and Removing Debris
- iii. Learners remove roots, stones, and stumps that may hinder planting.



Activity 3: Primary Tillage

- i. Learners dig or plough the soil using hoes to loosen the soil.
- ii. Leveling the Land

- iii. Use rakes to level the soil surface to make planting easier.



Activity 4: Marking Rows

- i. Using pegs, string, and measuring tape, learners mark rows according to recommended maize spacing (e.g., 75 cm between rows and 30cm within rows).

Observation and Discussion

- i. Each group presents their prepared section and explains the steps they followed.



Activity 5: Sowing, Planting, Plant Depth, Spacing, irrigation

Learner Materials Needed: Maize seed, Hoes, Measuring tape, Pegs and string, Watering cans (if necessary)



Activity 1: Sowing, Planting, Plant Depth, Spacing, irrigation

Procedure:

- i. Divide learners into small groups of 4–5.
- ii. Learners use pegs, string, and measuring tape to mark rows with recommended maize spacing (e.g., 75 cm between rows and 25–30 cm between planting stations).
- iii. Using hoes, learners dig small holes about 3–5 cm deep along the marked rows.

- iv. Learner's place 1–2 maize seeds in each hole.
- v. Learners gently cover the seeds with soil and press lightly.
- vi. Learner's water the planted seeds.
- vii. Learners observe the planted field and discuss why spacing and planting depth are important.



Content Tip

The teacher explains the importance of correct maize planting methods such as proper spacing, planting depth, and number of seeds per station.



Activity 6 – Weeding and fertilizer application

Learning Materials Needed: Hoes, Fertiliser, (e.g., basal or top-dressing fertiliser), Measuring cups or bottle tops, Gloves, Maize field or garden plot

Part 1: Weeding Activity

Procedure:

- i. The teacher takes learners to the maize field and asks them to identify weeds growing among maize plants.
- ii. Learners discuss why weeds are harmful to maize (competition for nutrients, water, and sunlight).
- iii. Divide learners into small groups of 4 - 5.
- iv. Each group weeds a small section of the maize field using hoes.
- v. Learners ensure they remove weeds carefully without damaging maize plants.

- vi. Collected weeds are placed in a pile for composting or disposal.

Discussion Questions:

Why is weeding important in maize production?

When should weeding be done in maize fields?

Part 2: Fertiliser Application Activity

Procedure:

- i. The teacher explains the types of fertilisers used in maize production (basal and top-dressing).
- ii. Demonstrate the ring or side-dressing method of fertiliser application.
- iii. Learners measure a small amount of fertiliser using a bottle top or measuring cup.
- iv. Each learner applies fertiliser about 5–10 cm away from the maize plant.
- v. Learners lightly cover the fertiliser with soil to prevent nutrient loss.



Activity 7: Harvesting Maize

Learner Materials Needed: Mature maize field or maize plants, Baskets or sacks

Knives (optional), Gloves

Procedure:

Introduction

The teacher explains the importance of harvesting maize at the correct stage of maturity.

Identifying Mature Maize

Learners observe maize plants and identify signs of maturity such as:

- Dry yellow/brown leaves
- Hard grains
- Dry husks covering the cob

Group Formation and harvesting

- i. Divide learners into small groups 4 - 5.
- ii. Harvesting the Maize
- iii. Learners harvest maize cobs by twisting and pulling the cob downward from the stalk.
- iv. Alternatively, learners may use knives carefully if necessary.
- v. Learner's place harvested maize cobs into baskets or sacks.

Sorting

- i. Learners separate good cobs from damaged or diseased ones.



Discussion

- i. Learners discuss the importance of harvesting at the correct time and proper handling of maize after harvesting.

5.2.6 Adding value, storing and marketing cereal crops.

Learning Materials: Samples of cereal crops, Grinding equipment, Sacks, plastic buckets, Weighing scale, Charts, Record sheets, Marker pens, Flip charts



Activity: Adding value, storing and marketing cereal crops

Procedure

- i. Divide learners into small groups of 4–5.
- ii. Provide each group with samples of cereals, for example maize.
- iii. Ask learners to identify the cereals and let them discuss possible ways of adding value.
- iv. Ask learners to demonstrate a simple value-addition process, for example grinding or pounding cereal.
- v. Ask learners to show how the cereal or processed product can be stored.
- vi. Ask learners to discuss and suggest ways of packaging and selling the products.
- vii. Ask each group to present their ideas on value addition, storage and marketing to the class.

Supporting Activities

- i. Participating in agricultural field days, shows and exhibitions.
- ii. Consulting print and digital resources for innovative ideas.

5.2.7Analysing yields and quality of the produce for possible improvement.

Learning materials: Samples of harvested crops, weighing scale, record book, moisture meter, measuring tape or rulers, calculator, charts/tables for yield comparison.



Activity 1: Analysing yields and quality of the produce for possible improvement.

Procedure

- i. Divide learners into small groups of about 4–5.
- ii. Provide crop samples and learning materials such as weighing scales, record sheets and measuring tools.
- iii. Ask learners to determine and measure the yield and quality of the provided samples (weight, size, colour, moisture and damage).
- iv. Ask learners to record their findings on data sheets provided.
- v. Ask each group to analyse and compare results to identify factors that may affect yield and quality.
- vi. Allow learners to suggest possible improvements.
- vii. Each group presents their findings to the class for discussion and feedback.



Content Tip

- Measuring and recording crop yield.
- Identifying factors that affect crop yield and quality.
- Suggesting ways to improve field crop production.

Supporting Activities

- Field visits.
- Sample sorting – learners sort harvested crops based on size, colour and quality.
- Viewing video clips
- Consulting electronic or printed media

5.2.8 Exploring innovative ways of increasing productivity.

Learning materials: Soil testing kit, fertilisers and organic manure, improved seed varieties, irrigation tools, pest and disease identification guides, growth charts and record sheets.



Activity 1: Exploring innovative ways of increasing productivity.

Procedure

- i. Divide learners into small groups of 4–5.
- ii. Ask learners to select certified seeds, fertilizers, irrigation tools and pest guides.
- iii. Ask each group to apply one or more methods to boost crop growth on small plots in the school garden.
- iv. Let learners observe over time.
- v. Learners keep records of crop growth, health, and yield.
- vi. Ask learners to analyse results.
- vii. In their groups they present their findings and discuss practical applications.



Content Tips

- Learners should consider factors of production most likely to improve the crop yields.
- Understanding different innovative methods to increase crop yield/productivity.
- Observing and recording the effects of these methods on crop growth and yield.
- Analysing and comparing results to identify the most effective production practice.

Supportive Activities

- Consulting print and digital resources



Skills Developed

- **Communicating** – Learners working in groups sharing scientific findings with their peers on analysing yields and quality of produce.
- **Collaborating** – Learners working together to achieve assigned tasks on adding value, storing and marketing cereal crops.
- **Analytically Thinking** – Learners are able to determine plant populations using specific
 - examples.
- **Problem Solving** – Learners exploring innovative ways of improving or increasing productivity to secure production.
- **Environmental Sustainability** – Learners exploring crop rotation plans to maintain soil fertility

Expected Standard

Cereal crops grown accordingly



Assessment

Task 1

In a rural community, farmers are focusing on improving cereal crop production to ensure food security and generate income. Locally grown cereals such as maize, sorghum, and millet are widely cultivated. Farmers are seeking ways to optimize production through improved practices from land preparation to marketing. As a learner with knowledge of field crop production and farm management, you are invited to prepare a practical and analytical report for a community agricultural workshop.

1. Identify locally grown cereal crops in your area and describe their importance;
2. Select a suitable site for cereal crop production, explaining the factors considered in your selection;
3. Design a simple crop rotation plan that includes at least one cereal crop and one legume crop;
4. Determine the appropriate plant population and assess seed germination rates for your selected cereal crop;
5. Explain how to manage maize or any other locally grown cereal crop from planting to harvesting, including practices such as weeding, fertilization, and pest control;
6. Describe methods of adding value, storing, and marketing the harvested cereal crop, and
7. Analyze crop yields and suggest innovative ways to increase productivity on the farm.

Task 2

A farmer in Nakuyewa community has harvested a large quantity of maize and sorghum. Some of the grain is getting spoiled because it is not properly stored, and the farmer is also selling the cereals at a very low price without processing them. As a learner of Agricultural Science, explain how you would help the farmer by:

- i. Suggesting two ways of adding value to the cereal crops;
- ii. Describing two proper storage methods to prevent damage and losses, and
- iii. Explaining two ways the farmer can market the cereal products to earn more income.



Sub-Topic 2.5.2 Summary

Field crop production is the process of growing crops such as cereals, legumes and oil crops on a large scale in open fields. It involves activities like land preparation, planting, fertilization, pest control and harvesting to obtain good yield and quality produce. Proper management practices help increase productivity and ensure food security.



Topic 2.5 Summary

This topic focused on crop production. Learners demonstrated an understanding of the external and internal morphology of plants and acquired practical skills in cultivating cereal crops, from planting to harvesting. They also explored methods for adding value to produce, as well as techniques for storage and marketing cereal crops. Additionally, learners analysed yields and assessed the quality of the produce to identify areas for improvement, while investigating innovative approaches to enhance productivity and efficiency in crop production.

TOPIC 2.6.0 LIVESTOCK PRODUCTION



Introduction

This topic will focus on poultry production in Zambia. Learners will focus on demonstrating understanding of poultry production, anatomy and physiology for poultry livestock and rearing of broilers from day-old to maturity.



General Competences

- Analytical thinking,
- Communication,
- Creativity and innovation,
- Collaboration,
- Critical thinking,
- Environmental sustainability and
- Problem solving.

Hook: *“Do you know that good management of livestock improves farm productivity?”*



Key Concepts / Processes / Terms

Breeding - mating animals to produce offspring with desired traits.

Broiler- a chicken raised mainly for meat production

Brooding- providing heat and care for young chicks

Biosecurity - measures used to prevent disease from entering or leaving a farm.

Cock/rooster- an adult male chicken

Candling - checking eggs with light to see if they are fertilized.

Culling - removing unproductive or sick animals from the herd or flock.

Hen- an adult female chicken

Incubation- the process of keeping eggs under the right temperature and humidity until they hatch

Poultry - domesticated birds raised for meat, eggs, furthers

Ration - the amount of feed given to an animal in 24 hours.

Vaccination - giving vaccines to prevent disease

Sub-Topic 2.6.1 Poultry Production in Zambia



Introduction

In this sub-topic, learners will explore the principles and practices of poultry production. They will develop an understanding of the anatomy and physiology of poultry and apply this knowledge in practical activities, including rearing broiler chickens to maturity while observing appropriate management practices such as feeding, housing, health care, and hygiene. In addition, learners will examine methods of adding value to poultry products and explore effective marketing strategies. They will also analyse yields and assess the quality of poultry products with the aim of identifying areas of improvement. Furthermore, learners will be encouraged to investigate innovative approaches that can enhance productivity and efficiency in poultry production.

Specific Competences

1. Demonstrate understanding of poultry production
2. Demonstrate understanding of anatomy and physiology for poultry livestock
3. Rear broiler chickens to maturity

Specific Competence 1

- Demonstrate understanding of poultry production

Learning Activities

The following are some of the activities you can use to teach the learners about poultry production.

6.1.1 Describing poultry.

Learning materials: Live chickens, ducks, turkeys, or pictures/models, charts



Activity: Describing poultry.

Procedure

- i. Put learners into small groups of 4-5.
- ii. Ask learners to describe the term poultry.
- iii. Ask learners to observe carefully the physical features of a live poultry e.g., chickens.
- iv. Ask learners to record their observations on a chart.
- v. Ask each group to present their findings to the class.
- vi. Ask learners to take note of the class findings.



Content Tip

Learners focus on physical features i.e., size, colour, comb type, feather pattern, beak, and leg colour.

Supporting Activities

- i. Viewing video clips
- ii. Consulting electronic or printed media

6.1.1 Describing poultry.



Activity 2: Field visit

Procedure

- i. Put learners into small groups of 4-5.
- ii. Ask learners to visit a local poultry farm.
- iii. Ask learners to observe different poultry breeds and management systems.
- iv. Ask learners to observe and record the following: breed, age, colour, egg production, feed type, and behaviour.
- v. Ask learners to present a descriptive report in their groups.
- vi. Ask learners to record the findings from the presentations.



Content Tip

- Learners to apply descriptive skills in real-life settings.
- Learners are evaluated on accuracy, completeness, and use of appropriate terminology
- Correct classification and justification demonstrate competency.

Supporting Activities

- i. Viewing video clips or pictures.
- ii. Consulting other digital or print media.
- iii. Participating in agricultural field days, shows and exhibitions.

6.1.2 Investigating the importance of poultry production in Zambia

Learning materials: Notebook, charts, manilla paper



Activity 1: Investigating the importance of poultry production in Zambia

Procedure

- i. Put learners into small groups of 4-5.
- ii. Ask each group to research on how poultry contributes to the Zambian economy.
- iii. Ask learners to prepare a short report or presentation summarizing their findings.
- iv. Ask each group to present their findings to the class.



Content Tip

- Importance of poultry farming (income generation, employment, trade, raw materials...)
- Learners identify the economic importance of poultry production in Zambia.

6.1.3 Categorizing chickens according to breeds and usage (layers, boilers, duals)

Learning materials: Different breeds of live chickens (Rhode Island Red, White Leghorn, Cornish Cross), Pictures or models of common breeds of chickens.



Activity1: Categorizing chickens according to breeds and usage (layers, boilers, duals)

Procedure

- i. Put learners into small groups of 3-5.
- ii. Ask learners to observe physical features of live chickens provided.
- iii. Ask learners to record observations in a table indicating breed, purpose, and characteristics.
- iv. Ask each group to present their findings in class.



Content tip

- Accuracy in identifying breed characteristics and correctly linking them to usage
- Provide learners with live chickens, pictures, or models of common breeds (e.g., Rhode Island Red, White Leghorn, Cornish Cross).
- Learners observe physical features: body size, feather color, comb type, and other distinguishing traits.

Supporting Activities

- i. Participating in agricultural field days, shows and exhibitions.
- ii. Consulting print and digital resources
- iii. Arrange a visit to a local poultry farm.

6.1.4 Exploring poultry hatcheries in Zambia

Learning Materials Needed: Internet access or printed information, Map of Zambia, Flip charts or notebooks, Markers/pens



Activity 1: Exploring poultry hatcheries in Zambia

Procedure

- i. Divide learners into small groups of 3 – 5
- ii. Assign each group one poultry hatchery in Zambia to explore, for example:
Hybrid Poultry Farm, Ross Breeders Zambia, Zamhatch, Panda Hatchery
- iii. Each group researches and records the following information:
 - Name of the hatchery
 - Location in Zambia
 - Type of chicks produced (broilers, layers, etc.)
 - Importance of the hatchery to farmers
 - Services provided (sale of chicks, vaccines, feed advice)
- iv. Groups present their findings to the class.



Discussion

- i. Learners mark the hatchery locations on a map of Zambia.
- ii. Why are hatcheries important to poultry farmers
- iii. How can hatcheries help improve poultry production in Zambia.
- iv. Learners write a short paragraph explaining the role of poultry hatcheries in the poultry industry in Zambia.



Content Tip

- Learning should be able to identify and describe some poultry hatcheries found in Zambia and explain their role in poultry production.

6.1.5 Exploring different systems of poultry rearing in Zambia (intensive-battery cage and deep litter, semi-intensive-fold, extensive-free range)

Learning Materials Needed: Pictures or charts of poultry housing systems, Flip charts or manila paper, Markers, Notebook and pens



Activity 1: Exploring different systems of poultry rearing in Zambia (intensive-battery cage and deep litter, semi-intensive-fold, extensive-free range)

Procedure:

- Divide learners into three groups.
- Assign each group one poultry rearing system:
Group 1: Free range system
Group 2: Semi-intensive system
Group 3: Intensive system

Exploration Task

- Each group should discuss and write down:
- Description of the system

- iii. How birds are managed in that system
- iv. Advantages of the system
- v. Disadvantages of the system
- vi. Examples of where it is commonly practiced in Zambia
- vii. Each group presents their findings to the class.
- viii. Other learners ask questions or add ideas.
- ix. The teacher summarizes the key points and highlights the differences between the systems.



Discussion

Learners write a short summary comparing the three poultry rearing systems used in Zambia.



Content Tip

- The teacher should briefly explain in the introduction that poultry in Zambia can be reared using different systems such as: Free range system, Semi-intensive system and Intensive system (deep litter or battery cage)

6.1.6 Analysing the characteristics of a good suitable poultry house (leaked proof roof, damp proof concrete floor, proper ventilation, east-west orientation, strong walls, vermin free, footbath...)

Learning Materials: Pictures or diagrams of poultry houses, Chart paper or manila paper, Markers, Notebook and pens



Activity 1: Analysing the characteristics of a good suitable poultry house (leaked proof roof, damp proof concrete floor, proper ventilation, east-west orientation, strong walls, vermin free, footbath...)

Procedure

- i. Divide learners into small groups of 4-5.
- ii. Provide each group with a diagram or picture of a poultry house.
- iii. Ask learners to identify and discuss whether the house has good characteristics.
- iv. Each group should analyse and explain why the characteristics they have suggested are important.
- v. Each group presents their analysis to the class.
- vi. Other learners may add ideas or ask questions.



Discussion:

- i. Learners draw a well-designed poultry house and label the important features discussed in class.



Content Tip.

- Features of a good poultry house are; Leak-proof roof, Damp-proof structure, Concrete floor, Proper ventilation, East–west orientation, Strong walls, Vermin-proof design, Footbath at the entrance

Each group should analyse and explain:

- Why a leak-proof roof is important
- Why the house should be damp-proof
- Importance of a concrete floor
- Why proper ventilation is necessary
- Reason for east–west orientation
- Importance of strong walls

- How to keep the house vermin-free
- Role of a footbath in disease control

6.1.7 Identifying various poultry equipment (feeders, drinkers, heat sources, lighting sources, cages, perches...)

Learning Materials Needed: Real poultry equipment or pictures of equipment, Flash cards with names of equipment.



Activity1: Identifying various poultry equipment (feeders, drinkers, heat sources, lighting sources, cages, perches...)

Procedure:

- i. Divide learners into small groups of 3-5
- ii. Let the learners observe and record the equipment provided
- iii. Learners discuss the observed equipment in groups
- iv. Learners present their findings to the class



Content Tip

Learners should be able to identify common poultry equipment and explain their uses in poultry production.

6.1.8 Making various poultry equipment using local materials (drinkers, feeders, heat source, light source, cages, perches...)

Learning materials: Bottles, buckets, wood, bamboo, wire, rope, nails, etc.



Activity 1: Making various poultry equipment using local Materials

Procedure

- i. Form groups of 4–5 learners.
- ii. Ask learners to discuss the purpose of each equipment.
- iii. Gather local materials: bottles, buckets, wood, bamboo, wire, rope, nails, etc.
- iv. Ask learners to plan and design each equipment before starting.
- v. Ask learners to carefully construct the equipment:
 - o Drinkers and feeders: cut and attach bottles and buckets.
 - o Cages: use wood and wire mesh.
 - o Perches: fix bamboo or wooden poles.
 - o Heat and light: set up safely.
- vi. Test each equipment for stability and proper function.
- vii. Ask learners to share and discuss how well the equipment works.



Content Tips

- Handle tools and materials safely.
- Use local materials creatively.
- Make equipment strong, stable, and suitable for birds.
- Work together and keep the workspace clean.
- Check and discuss how well the equipment works.

Supporting Activities

- i. Participating in agricultural field days, shows and exhibitions.
- ii. Consulting print and digital resources

6.1.9 Construct a Simple Poultry House Using Locally Available Materials

Learning materials: Wooden poles or timber, Grass or thatch, Clay bricks or mud, Bamboo or reeds, Wire mesh or chicken wire, Sand and stones, Nails, Wooden planks



Activity 1: Construct a simple poultry house using locally available materials

Procedure

- i. Divide learners into small groups of 4–5.
- ii. Ask each group to identify and collect available materials.
- iii. Ask learners to select a suitable site for construction.
- iv. Instruct learners to measure and mark the layout.
- v. Ask learners in their groups to fix the poultry foundation.
- vi. Learners should construct the poultry house using the specifications if any.
- vii. Ask learners to present their constructed poultry house.
- viii. Learners should explain in groups how they constructed it and the materials used.



Content Tips

- Choose a suitable site.

- Mark the poultry house layout on the ground.
- Fix wooden poles in the ground and construct the frame, then finally the roof.

Supporting Activities

- i. Participating in agricultural field days, shows and exhibitions.
- ii. Consulting print and digital resources for innovative ideas



Skills Developed

- **Communicating** – Learners explaining poultry management practices to peers
- **Collaborating** – Learners sharing responsibilities such as vaccinations and cleaning of the poultry house.
- **Analytically Thinking** – Learners comparing weight gain or production at different stages
- **Problem Solving** – Learners identifying problems such as diseases, overcrowding or poor feeding
- **Environmental Sustainability** – Learners using locally available materials and resources responsibly
- **Creating and Innovating** – Learners designing simple feeders, drinkers or poultry houses using local materials

Expected Standard

Understanding of poultry production demonstrated appropriately.



Assessment

Task 1

In Zambia, poultry farming is a key component of livestock production that contributes to household nutrition, income generation, and employment. Communities are increasingly adopting poultry rearing as a business, and learners are expected to acquire practical and theoretical knowledge of poultry production. As a learner with knowledge of poultry production, management, and farm construction, you are invited to prepare a presentation and practical demonstration for a school-based poultry awareness program.

- i. Describe poultry and explain the importance of poultry production in Zambia;
- ii. Categorise chickens according to their breeds and usage (e.g., meat, eggs, dual purpose);
- iii. Explore some poultry hatcheries and explain their role in poultry production;
- iv. Describe different systems of poultry rearing (e.g., intensive, semi-intensive, free-range);
- v. Analyse the characteristics of a good poultry house suitable for local conditions;
- vi. Identify various poultry equipment commonly used on a farm;
- vii. Demonstrate how to make selected poultry equipment using locally available materials, and
- viii. Construct a simple poultry house using local materials, ensuring it meets basic requirements for ventilation, protection, and hygiene.

Task 2

A farmer in a village of Shangombo keeps goats and chickens, but the animals are not producing well. Many chickens lay few eggs, some goats are often sick, and the animals look thin. The farmer also keeps them in a poorly ventilated shelter and feeds them irregularly.

- i. Discuss the situation and suggest ways the farmer can improve livestock production through good management, and
- ii. Identify the problems affecting livestock production on this farm.

Specific Competence 2

Demonstrate understanding of anatomy and physiology for poultry livestock

Learning Activities:

The following are the activities you can use to teach learners about anatomy and physiology for poultry livestock

6.2.1 Exploring the external structures of chickens

Learning materials: Live chicken or picture/chart of a chicken, notebook and pencil



Activity 1: Exploring the external structures of chickens

Procedure

- i. Put learners into small groups of 4-5.
- ii. Ask learners to observe a chicken carefully by looking at different external body parts.
- iii. Ask learners to draw the chicken and label the parts on flip charts or manilla paper
- iv. Ask learners to discuss the function of all the external parts of a chicken
- v. Ask learners to present their findings



Content Tip

- The following external parts of the chicken to be identified during group discussion i.e., beak, comb, wattles, wings, feathers, eyes, legs, claws, tail

6.2.2 Exploring the external structures of chickens

Learning materials: Clay, paper, cardboard, notebook and pencil



Activity 2: Model Making

Procedure:

- i. Put learners into small group of 4-5
- ii. Ask learners to create a chicken model using available local materials
- iii. Ask learners to label its external structures state their functions.
- iv. Ask learners to present their findings



Discussion

Explain the functions of the external parts of a chicken structure.

Supporting Activity

- i. Consulting print and digital resources

6.2.3 Exploring parts of the digestive system in a dissected chicken

Learning Materials: A chicken, gloves, knife, dissection tray, chart/diagram of a chicken digestive system, notebook and pencils



Activity 1: Exploring parts of the digestive system in a dissected chicken

Procedure

- i. Put learners into small groups of 4-5
- ii. Ask learners to dissect a chicken to examine the digestive system
- iii. Ask learners to carefully observe and record all the internal organs of the digestive system of a chicken
- iv. Ask learners to draw the digestive system as observed in the dissected chicken
- v. Ask learners to present their findings



Discussion

- i. Which organ stores food before digestion?
- ii. Which organ grinds food?
- iii. Why is the gizzard strong and muscular?



Content Tips

- The teacher guides learners to locate the digestive organs i.e, Beak/mouth, oesophagus, crop, proventriculus (glandular stomach), gizzard, small intestine, large intestine, cloaca
- Sequence: Mouth → Oesophagus → Crop → Proventriculus → Gizzard → Small Intestine → Large Intestine → Cloaca

Supporting Activities

- i. Consulting print and digital resources

6.2.4 Illustrating the parts of the digestive system of a chicken in relation to their functions

Learning Materials: A chicken, gloves, knife, dissection tray, chart/diagram of a chicken digestive system, notebook and pencils, charts or pictures of the digestive system



Activity 1: Illustrating the parts of the digestive system of a chicken in relation to their functions

Procedure:

- i. Put learners into small groups of 4-5
- ii. Ask learners to discuss the following questions;
 - a. What do chickens eat?
 - b. How does food move inside a chicken's body?
- iii. Ask learners to dissect a chicken to examine the digestive system.

- iv. Ask learners to carefully observe the digestive system of a chicken.
- v. Ask learners to discuss the parts of the digestive system of a chicken in relation to their functions.
- vi. Ask learners to record suggested points and present their findings.

6.2.5 Examining the parts of the reproductive system in a dissected chicken

Learning materials: A chicken (hen), dissection tray, gloves, diagram/chart of a chicken reproductive system, notebook and pencil.



Activity 1: Examining the parts of the reproductive system in a dissected chicken

Procedure

- i. Put learners into small groups of 4-5
- ii. Ask learners to dissect a chicken to examine the reproductive system
- iii. Ask learners to carefully observe and record all the internal parts of the reproductive system of a chicken
- iv. Ask learners to draw parts of the reproductive system as observed in the dissected chicken
- v. Ask learners to present their findings



Content Tip

- In a hen, the following reproductive organs should be identified, Ovary, oviduct, infundibulum, magnum, isthmus, uterus (shell gland), cloaca
- In a cock, following reproductive organs should be identified, testes, vas deferens

Supporting Activities

- i. Viewing video clips

- ii. Consulting electronic or printed media

6.2.5 Illustrating the parts of the reproductive system of a hen in relation to their functions

Learning Materials: Chart, model of hen reproductive system, pictures, specimen of reproductive system (actual), dissecting tools (scalpel/knife, scissors), gloves, projectors



Activity 1: Illustrating the parts of the reproductive system of a hen in relation to their functions

Procedure

- i. Divide learners into small groups of 4–5.
- ii. Ask learners in groups to obtain charts, diagrams, or a specimen of a hen's reproductive system.
- iii. Ask learners to observe the materials carefully.
- iv. Ask learners to identify the main parts of the reproductive system.
- v. Ask learners to discuss the functions of each part in their groups.
- vi. Ask learners to draw the reproductive system on manila paper or in their books.
- vii. Each group should explain their diagrams to the class.



Discussion

Describe the functions of the reproductive system of a hen



Content Tip

- The mature hen reproductive parts include: ovary, infundibulum, magnum, isthmus, uterus, vagina, uterus is also called shell gland.

Supporting Activities

- i. Viewing video clips of the reproductive system of a hen in relation to their functions
- ii. Consulting electronic or printed media

6.1.6 Relating the structure to the function of the parts of an egg.

Learning Materials: Fresh eggs, diagrams/charts, magnifying glass, knife, tray, gloves, apron, manila paper, markers/pencils, textbooks



Activity 1: Relating the structure to the function of the parts of an egg.

Procedure

- i. Divide learners into small groups of 4–5.
- ii. Ask learners to observe the external parts of the egg.
- iii. Ask learners to observe internal parts of the egg.
- iv. Ask learners to identify and discuss each part.

- v. Ask learners to use charts to match each part to its function.
- vi. Ask each group to draw and label the egg showing structure and function.
- vii. Learners to make presentations in groups.



Content Tips

- i. Careful breaking of the egg is needed when observing the internal parts.
- ii. Internal parts include yolk, albumen, chalaza, air space.

Supporting activities

- i. Viewing video clips on the internal structure of an egg
- ii. Consulting electronic or printed media



Skills developed

- **Communication skill** –Learners presenting findings and explaining functions to the class.
- **Collaboration/Teamwork** – Learners working effectively in small groups.
- **Critical Thinking** – Discussing why each part of the egg is important for reproduction.
- **Analytical skills** – Through relating the structure of each part to its function.
- **Problem Solving skill** – Identifying structures correctly and linking them to their roles.

- **Environmental Sustainability** – Learners understanding how efficient feeding and digestion reduces feed wastage and environmental impact.
- **Creating and Innovating**-Learners think of ways to improve feeding systems or breeding management for better productivity.

Expected Standard

Understanding of the anatomy and physiology of chickens demonstrated correctly.



Assessment

Task 1

At an agricultural school, learners are participating in a poultry science practical project aimed at improving understanding of chicken anatomy, physiology, and reproduction. The goal is to equip learners with knowledge and skills to manage poultry for optimal production and productivity. As a learner with knowledge of poultry anatomy, digestive and reproductive systems, and egg structure, you are required to prepare a comprehensive practical report and presentation for your class and community stakeholders.

- i. Explore and describe the external structures of a chicken, highlighting features such as the beak, comb, wattles, wings, feathers, legs, and tail;
- ii. Identify and examine the parts of the digestive system in a dissected chicken, including the beak, oesophagus, crop, proventriculus, gizzard, small intestine, large intestine, and cloaca;
- iii. Illustrate the digestive system of a chicken and explain the function of each part in the digestion process;
- iv. Examine the parts of the reproductive system in a dissected chicken (male and female), including ovary, oviduct, testes, vas deferens, and cloaca;

- v. Describe the parts of the reproductive system of a hen in relation to their functions, including ovary, infundibulum, magnum, isthmus, uterus (shell gland), vagina, and cloaca, and
- vi. Relate the structure of an egg (shell, albumen, yolk, chalaza, air space) to its function in protecting and nourishing the developing embryo.

Task 2

You are a farm manager at a poultry farm in Chavuma district. Lately, some hens have been laying fewer eggs, and some eggs are malformed. You need to investigate the problem and suggest ways to improve egg production.

- i. List the main parts of the hen reproductive system;
- ii. Explain the functions of each part;
- iii. Suggest how problems in these parts could affect egg production;
- iv. Recommend ways to improve egg production, and
- v. Draw and label the reproductive system showing each part's role.

Specific Competence 3

Rear broiler chickens to maturity

Learning Activities

The following are some of the activities you can use to teach learners about rearing broiler chickens from day-old to market weight.

6.1.1 Managing broiler chickens from day-old to market weight

Learning Materials: Chart paper or notebooks, Pens and markers, Broiler production guide, calculator



Activity: Managing broiler chickens from day-old to market weight

Procedure:

- i. Divide learners into small groups of 3-5
- ii. Each group is asked to develop a broiler production plan from day-old chicks up to market weight (6-8 weeks)
- iii. Learners should include the following in their plan;
 - Preparation of the poultry house before chicks arrive,
 - Brooding management (heat, bedding, light),
 - Feeding management (starter, grower and finisher),
 - Water provision,
 - Vaccination and disease control,
 - Routine management practices (cleaning, litter management, culling),
 - Growth monitoring and record keeping,
 - Marketing time and expected market weight.
- iv. Each group organizes the plan week by week in a table or chart
- v. Ask each group to present their plans to the class



Content tip

- Ensure that necessary calculations such as calculations to do with feed conversion ratio are referred to when planning.
- The teacher guides the discussion and corrects any mistakes

1.1.2 Preparing a Poultry House for the Arrival of Day-Old Chicks

Learning Materials: Brooms, disinfectant, water, litter materials (wood shavings or sawdust), feeders, drinkers, heat source (charcoal stove, briskets, gas brooder or electric bulb), footbath, thermometer.



Activity: Preparing a Poultry House for the Arrival of Day-Old Chicks

Procedure:

- i. Take learners to a poultry house.
- ii. Learners carry out the following tasks;
 - Clean the poultry house by removing old litter and dirt,
 - Wash and disinfect the floor, walls, feeders, and drinkers,
 - Spread dry litter material evenly on the floor,
 - Arrange feeders and drinkers properly in the house,
 - Install a heat source for brooding,
 - Place a footbath with disinfectant at the entrance,
 - Check ventilation and ensure the house is leak-proof and vermin-free,
 - Learners check that the house is warm and ready before chicks arrive.
- iii. Ask learners to discuss why each preparation step is important
- iv. Learners should present what they have discussed



Content Tip

- This activity should enable learners to understand and practice the correct preparation of a poultry house before introducing day-old chicks. Learners will be able to correctly prepare a poultry house to ensure a safe and healthy environment for day-old chicks.

6.1.3 Formulating feed for broilers

Learning Materials: Feed ingredients (Maize bran, Maize meal number 3, Soyabean Meal, Fishmeal, Meat and bone meal, Salt, dicalcium phosphate, broiler vitamin and mineral premix), broiler feed standards, spade, buckets or basins or clean floor.



Activity: Formulating feed for broilers

Procedure:

- i. Divide learners into 3 groups
- ii. Ask learners to calculate the proportions of ingredients required to make a simple broiler formula (Pearson Square method should be used to balance ingredients) example (Crushed maize 60%, Soya meal 25%, Fishmeal 10%, Lime 3%, Premix 2%)
- iii. Give each group ingredients required to make a particular broiler feed (starter, grower or finisher feed)
- iv. Ask the learners in groups thoroughly mix the ingredients
- v. Learners should observe the colour, texture and smell of the finished feed
- vi. Store mixed feed in clean bags or containers



Content Tip

- The teacher should emphasize calculating the right ratios of ingredients so that the nutrient levels in the formulated feed match the animal requirements.



Discussion question

- i. Learners explain why each ingredient is included in broiler feed
- ii. Discuss the importance of balanced nutrition

6.1.4 Carrying Out Routine Management Practices in Broiler Rearing

Learning Materials needed:

Broiler chickens, feeders, drinkers, broiler feed, clean water, litter material, disinfectant, record book, weighing scale.



Activity: Carrying Out Routine Management Practices in Broiler Rearing

Procedure:

- i. Divide learners into small groups and assign them to a broiler unit.
- ii. Place day old chicks in the brooder and manage them
- iii. Vaccinate birds following an approved vaccination schedule
- iv. Each group performs routine management practices such as:
- v. Feeding broiler chicken with the correct type of feed from day old to market weight
- vi. Providing clean drinking water throughout the rearing period
- vii. Cleaning and refilling feeders and drinkers.
- viii. Checking and replacing wet litter.
- ix. Observing birds for signs of disease or stress.
- x. Recording mortality and feed consumption.
- xi. Learners also check ventilation, temperature, and general cleanliness of the poultry house.
- xii. After the activity, each group reports the tasks performed and observations made.



Content Tip

- To enable learners to practice routine management activities required for proper growth and health of broiler chickens.
- Discussion Questions

6.1.5 Adding value and marketing (dressing, packaging, smoking of broiler chickens, storage and marketing)

Learning materials: Live or dressed broiler chickens, Knives and cutting boards, Weighing scale, Packaging materials, Labels and markers, Cooler box or refrigerator, Cleaning materials e.g. water, soap, gloves, Charts, Record books or work sheets, Market price information.



Activity 5. Adding Value and Marketing (dressing packaging, smoking of broiler chickens, storage, Marketing).

Procedure

- i. Divide learners into small groups of 4-5.
- ii. Provide each group with a slaughtered broiler chicken and the required materials (knives, cutting boards, and packaging materials, weighing scale ...)
- iii. Ask learners to wash and prepare chickens under hygiene conditions.
- iv. Ask learners to cut the chicken into different portions (breasts, thighs, wings, and drumsticks) to add value.
- v. Ask learners to weigh the different portions using a weighing scale.
- vi. Ask learners to package the chicken portions properly using plastic bags.
- vii. Ask learners to label the packages with product name and price.
- viii. Ask each group to calculate the possible selling price and expected profit.
- ix. Let learners present their products and marketing ideas to the class.
- x. Conclude the activity with a class discussion on value addition and good marketing in poultry production.



Content Tips

- Maintain good hygiene.
- Handle knives carefully.
- Cut the chicken into uniform portions.

- Use clean and suitable packaging.
- Label the products clearly.
- Consider market demand and prices.
- Store the processed chicken properly.

Supporting Activities

- Consulting digital resources and print media

6.1.6 Analysing yields and quality of the produce for possible improvement

Learning Materials: Live or slaughtered broiler chicken, Knives and cutting boards

Weighing scale, measuring chart/ record sheets, Protective clothing, Sample containers or trays, Calculator, Charts- showing quality grades of broiler meat, Pens, note books, Cleaning materials



Activity 6: Analysing yield and quality of the produce for possible improvement

Procedure

- i. Divide learners into small groups of 4–5.
- ii. Provide each group with a slaughtered broiler chicken and the required materials (knives, cutting boards, packaging materials, weighing scale, etc.).
- iii. Ask learners to observe the broiler chicken and record its general condition e.g. weight and appearance
- iv. Ask learners to weigh the live broiler chicken
- v. Slaughter and dress the chicken (ask learners to do this)
- vi. Weigh the dressed chicken-record the weight to determine the weight to determine the dressing percentage yield.
- vii. Ask learners to cut the chicken into parts such as breast, wings, thighs and drumsticks.

- viii. Ask learners to weigh the different parts and record the results for comparison.
- ix. Ask learners to observe the quality of the meat-noting factors such as color, tenderness, fat distribution and cleanliness.
- x. Ask learners to discuss within the groups, the factors that may affect yield and quality
- xi. Ask learners to present group findings to the class and suggest ways to improve broiler production and meat quality



Content Tip

- Factors that may affect yield and quality include:
- Breed – some breeds grow faster and produce more meat.
- Management – good feeding, vaccination and proper housing affect meat quality.
- Health – disease reduces body weight and value.
- Slaughter age – birds slaughtered at the correct age give good meat.
- Practice proper handling and hygiene.
- Use improved broiler breeds with high growth rate.
- Provide balanced and adequate feed to promote growth.
- Vaccinate and treat birds to control diseases.
- Maintain good housing, sanitation and ventilation

Supporting Activities

- i. Viewing video clips
- ii. Consulting electronic or printed media

6.1.7 Exploring innovative ways of increasing productivity.

Learning materials: Notebook, Feed stuffs, Beam balance, Chickens



Activity 1: Exploring innovative ways of increasing productivity.

Procedure

- i. Divide learners into small groups of 4-5.
- ii. Provide learning materials for each group.
- iii. Ask learners to observe the broiler house and let them identify the current management practices.
- iv. Ask learners to measure and record information such as body weight, amount of feed given and health condition of the birds.
- v. Ask learners to discuss within their groups possible innovative methods that can improve productivity.
- vi. Ask each group to suggest practical improvements that could increase growth rate, reduce mortality and improve feed efficiency.
- vii. Ask group to present their ideas to the class.
- viii. Ask learners to summarize the key innovations and record the findings.



Content Tips

- Using improved broiler breeds that grow faster and produce more meat
- Providing balanced and high- quality feed to promote faster growth.
- Using automatic or improved feeders and drinkers to reduce feed wastage.
- Maintaining proper temperature and Ventilation.
- Following proper vaccination, disease control, good hygiene, adequate space, proper lighting, use modern technology and improving brooding management.

Supporting activities

- i. Viewing video clips
- ii. Consulting electronic or printed media



Skills developed

- **Communicating** – Learners explaining broiler management practices such as feeding, housing, disease control.
- **Collaborating** – Learners cooperating when performing practical activities such as feeding, cleaning and record keeping
- **Analytically Thinking** – learners evaluating the quality and yield of poultry products for improvement
- **Problem Solving** – Learners making decisions to increase productivity and profitability
- **Environmental Sustainability** – learners promoting efficient use of feed, water and other resources
- **Creating and Innovating** – Learners developing new ways of adding value to poultry products e.g. packaging and processing

Expected standard

Broiler chickens reared to maturity satisfactorily.



Assessment

Task 1

A group of learners at Chingola Secondary School raised 100 broiler chickens for sale. After 6 weeks, they slaughtered the birds and discovered that some chickens were smaller in size, while others had good body weight and quality meat. The learners also realized that selling whole birds at the local market brought less profit than expected. Their teacher asked them to think of better ways to improve production, increase quality, and earn more money from the broilers.

As a group:

- i. Suggest two ways of adding value to the broiler chickens before selling them;
- ii. Explain two methods learners can use to analyze the yield and quality of the broiler meat;
- iii. Identify three possible reasons why some broilers had low body weight;
- iv. Suggest three innovative ways the learners can use to increase productivity in the next batch of broilers, and
- v. Propose two marketing strategies that can help the learners sell their broiler products at a better price.

Task 2

In a rural community, a poultry enterprise has been established to rear broiler chickens for meat production. The enterprise aims to train learners and farmers on modern poultry management practices to ensure high productivity, profitability, and quality meat supply. As a learner with knowledge of broiler production and farm management, you have been invited to prepare a practical report and presentation for a community poultry workshop.

- i. Explain how to manage broiler chickens from day-old chicks to market weight, including feeding, watering, health care, and environmental management;

- ii. Describe how to prepare a poultry house for the arrival of day-old chicks, including cleaning, disinfecting, bedding, and setting up feeders and drinkers;
- iii. Formulate appropriate feed for broiler chickens at different growth stages;
- iv. Carry out routine management practices such as feeding, watering, vaccination, and monitoring bird health;
- v. Explain methods of adding value and marketing broiler chickens, including packaging, pricing, and customer engagement;
- vi. Analyse yields and quality of the produce to identify areas for improvement, and
- vii. Explore innovative ways of increasing broiler productivity, such as improved housing, feed efficiency, and biosecurity measures.



Sub-Topic 2.6.1 Summary:

This topic focused on broiler production. Learners developed an understanding of the anatomy and physiology of poultry and applied this knowledge in the practical rearing of broiler chickens to maturity. Throughout the process, they considered and applied appropriate management practices, including proper feeding, housing, hygiene, and health care, to ensure successful broiler production.

Bibliography

1. East African Agriculture- D.N. Ngugi, P.K. Karau, W.Nguyo (1990) Macmillian London.
2. King AN (1985), Agriculture, An introduction for Southern African. London, Cambridge University Press.
3. Ian Elliott, Orville Wolsey (1995), Agricultural Science 1, A Junior Secondary Course for the Caribbean
4. G. Owen (1984), O-Level Agriculture, Longman Group Ltd, Singapore
5. Migwi J et al (2006), Golden Tips Agriculture, Kampala, Moran (E.A) Publisher.
6. Mwala .M, Mwangala .J (2017) “Vegetable production in Zambia”, University of Zambia Press.
7. Ngugi DN (1990), East African Agriculture, 3rd Ed, Macmillan Ltd Oxford.
8. Parsons BJ (19880, Agriculture Science for Senior Secondary, Ndola, Print Park Zambia.
9. Simon Hikaula (2015), Achievers Junior Secondary –EAEP Nairobi
10. W.A.Sakira, ‘O’ Level Agriculture, Principles and Practice, Oxford University Press Nairobi. Ministry of Education (2023)
Zambia Education Curriculum Framework CDC, Lusaka
11. Curriculum Development Centre. (2024). Agricultural Science Syllabus Form 1—4. ZEPH.